Supplementary Methods

Prepared by Sin Guan Kong

Supplementary Information Overview

The Supplementary Information was organized to provide a fully reproducible and conceptually structured description of the D-PHY framework, including transcriptomic preprocessing, pathway-level uncertainty construction, uncertainty-aware discriminability analysis, scale-dependent observation analysis, structured-versus-random validation, statistical reproducibility procedures, and conceptual interpretation under partial observability.

The supplementary materials are divided into eleven connected Supplementary Methods sections (SM1–SM11). These sections were designed to progressively guide the reader from raw transcriptomic preprocessing toward uncertainty-aware observable construction, discriminability analysis, random validation, cross-cancer integration, and reproducible figure generation.

The overall organization is summarized below.

Supplementary Methods Menu

SM1 | Overall analytical framework and computational workflow

Introduces the overall D-PHY analytical architecture, including pathway-level uncertainty representation, uncertainty-aware discriminability scoring, scale-dependent observation analysis, and structured-versus-random validation.

SM2 | TCGA transcriptomic cohort preparation and preprocessing

Describes TCGA transcriptomic loading, barcode harmonization, survival-table parsing, z-score normalization, quality-control procedures, and cohort-level preprocessing workflows.

SM3 | Pathway database construction and pathway-level observable generation

Describes GMT parsing, pathway mapping, pathway-level activity calculation, FWHM-based uncertainty-width estimation, and construction of pathway-level observable matrices.

SM4 | Observable coherence, uncertainty-aware discriminability, and λ-scan analysis

Defines pathway coherence, patient-level uncertainty-aware scoring, λ-dependent uncertainty penalization, Kaplan–Meier survival analysis, and clinical discriminability computation.

SM5 | Scale-dependent observation analysis and K-scale discriminability framework

Describes the K-scale observational framework, pathway ranking, scale-dependent discriminability analysis, optimal observation-scale identification, and K–λ interaction analysis.

SM6 | Structured-versus-random validation framework and random observable analysis

Describes the size-matched random-gene-set validation workflow, high-discriminability random-set dissection, and cross-cancer recurrence analysis.

SM7 | Statistical analysis, visualization workflow, and reproducibility framework

Summarizes statistical procedures, numerical stabilization methods, survival-analysis safeguards, visualization standards, and reproducibility constraints.

SM8 | Figure generation, data integration workflow, and manuscript-specific output construction

Describes the generation of manuscript figures, cross-cancer integration workflows, summary-table construction, and publication-ready visualization pipelines.

SM9 | Biological interpretation framework, partial observability formulation, and uncertainty-geometry representation

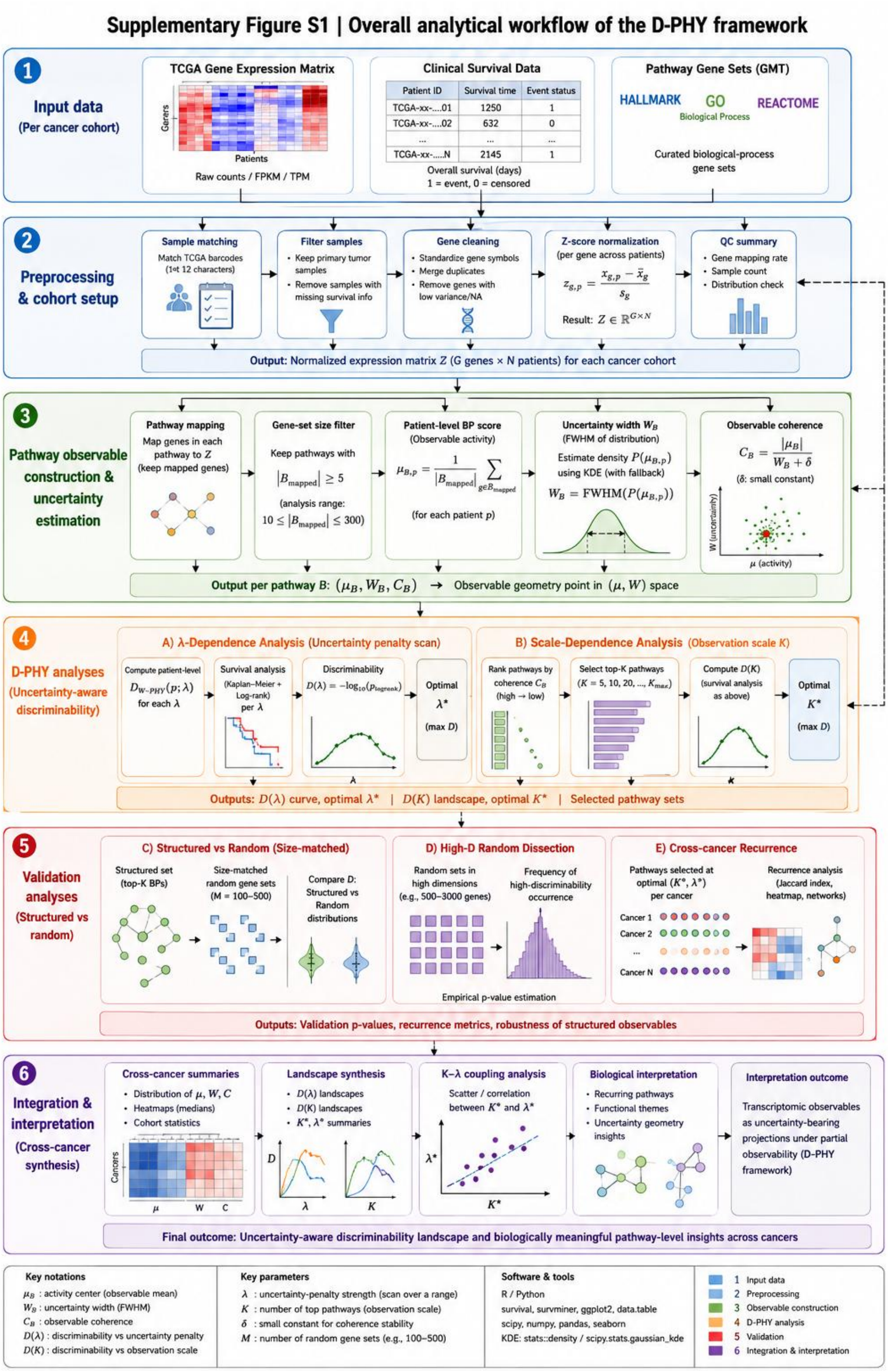
Provides the conceptual interpretation framework underlying D-PHY, including partial observability, pathway-level uncertainty geometry, state-cloud representation, and uncertainty-aware observable interpretation.

SM10 | Result interpretation framework, ranking strategy, evidence hierarchy, and methodological limitations

Defines the interpretation boundaries of discriminability, uncertainty width, coherence, λ, K-scale analysis, evidence hierarchy, exploratory analyses, and methodological limitations.

SM11 | Computational reproducibility framework, execution protocol, code organization, and data availability

Describes execution order, directory structure, script organization, output architecture, figure-regeneration procedures, reproducibility safeguards, and data/code availability.



Supplementary Figure S1 | Overall analytical workflow of the D-PHY framework

Supplementary Figure S1 | Overall analytical workflow of the D-PHY framework for uncertainty-aware discriminability analysis of pathway-level transcriptomic observables.

The analytical workflow begins with transcriptomic expression matrices, clinical survival annotations, and curated biological-process gene-set databases (Hallmark, GO:BP, and Reactome) collected independently for each cancer cohort. After cohort-specific preprocessing, including barcode matching, sample filtering, gene cleaning, and z-score normalization, pathway-level observables are constructed by mapping genes onto curated biological-process gene sets.

For each pathway B and patient p, pathway activity $\mu_{B,p}$ is computed as the mean z-scored expression across mapped genes, while uncertainty width W_B is estimated using FWHM-based distribution analysis. Observable coherence is subsequently calculated as $C_B = |\mu_B|/(W_B + \delta)$, generating pathway-level observable geometry in (μ, W) space.

The D-PHY framework then performs two complementary analyses: (i) uncertainty-penalty optimization through λ-dependent discriminability analysis, and (ii) scale-dependent observation analysis using K-scale pathway selection. Structured-versus-random validation is subsequently performed using size-matched random gene sets, high-discriminability random dissection, and cross-cancer recurrence analysis to evaluate the robustness and biological organization of pathway-level observables.

Finally, cross-cancer integration analyses summarize observable geometry, discriminability landscapes, λ–K coupling structure, and recurrent pathway organization across cancers. Under this framework, transcriptomic pathway observables are interpreted as uncertainty-bearing projections of latent biological organization under partial observability rather than direct measurements of unique latent states.

Supplementary Methods S1 | Overall analytical framework and computational workflow

This study developed an uncertainty-aware discriminability framework to evaluate transcriptomic biological-process observables in partially observable cancer systems. The central goal was to determine whether biological-process-level observables contain not only activity information, but also measurable uncertainty structure that influences their ability to separate clinical outcomes.

The complete analysis was organized into three connected modules:

1. **BP uncertainty-structure analysis**
2. **Uncertainty-aware discriminability analysis**
3. **Structured-versus-random validation analysis**

Together, these modules support the main claim that transcriptomic observables should be interpreted as structured, uncertainty-bearing projections rather than direct readouts of latent biological states.

S1.1 Input data and cohort-level organization

For each cancer cohort, the analysis started from a TCGA gene-expression matrix and a matched clinical survival table. Molecular and clinical records were matched using standardized TCGA patient identifiers, typically the first 12 characters of the TCGA barcode.

Only primary tumor samples were retained where sample-type information was available. Samples without valid survival time or event-status information were excluded from survival-based discriminability analysis.

The main D-PHY I analysis was applied across the following cancer cohorts:

BLCA, BRCA, COAD, GBM, HNSC, KIRC,

LAML, LIHC, LUAD, LUSC, OV, PAAD,

PRAD, SARC, SKCM, STAD, THCA, UCEC

Each cancer cohort was processed independently. No expression values were pooled across cancers before normalization, BP scoring, uncertainty estimation, or survival analysis.

This cohort-wise design was used to avoid cross-cancer normalization artifacts and to preserve cancer-specific observable structure.

S1.2 Gene-expression preprocessing

For each cohort, the gene-expression matrix was first cleaned and standardized.

Gene symbols were normalized to a consistent format, duplicated gene entries were merged where necessary, and patient columns were matched to survival records.

Each gene was then z-score normalized across patients within the same cancer cohort:

$$z_{g,p} = \frac{x_{g,p} - \bar{x}_g}{s_g}$$

where:

- $x_{g,p}$ is the expression value of gene g in patient p,
- $\bar{x}_g$ is the mean expression of gene g across patients,
- s_g is the standard deviation of gene g across patients,
- $z_{g,p}$ is the standardized expression value.

This produced a normalized expression matrix:

$$Z \in \mathbb{R}^{G \times N}$$

where G is the number of genes and N is the number of patients.

The z-score normalization step ensured that genes with different absolute expression ranges could be combined within BP gene sets without dominance by high-expression genes.

S1.3 Biological-process observable construction

Curated biological-process gene sets were used to construct transcriptomic observables. The main structured gene-set databases used in D-PHY I were:

Hallmark

GO Biological Process

Reactome

For each gene set B, only genes present in the normalized expression matrix were retained. Gene sets with too few mapped genes were removed from analysis.

The minimum mapped gene threshold was:

$$| B_{\text{mapped}} | \geq 5$$

For controlled analyses, an additional gene-set size filter was used:

$$10 \leq |B_{\text{mapped}}| \leq 300$$

This controlled range was used to reduce instability from very small gene sets and dilution from excessively broad gene sets.

For each patient p, the biological-process activity score was computed as the mean z-scored expression across mapped genes:

$$\mu_{B,p} = \frac{1}{|B_{\text{mapped}}|} \sum_{g \in B_{\text{mapped}}} z_{g,p}$$

This patient-level BP score represents the observed activity of biological process Bin patient p.

S1.4 Construction of BP-level uncertainty observables

After computing patient-level BP activity scores, each BP observable was summarized by two quantities:

1. activity magnitude, μ_B
2. uncertainty width, W_B

The cohort-level activity magnitude was computed from the patient-level BP score distribution. In operational terms, μ_B represents the central activity level or observable center of the BP.

The uncertainty width $W_B$was estimated using full width at half maximum:

$$W_B = \text{FWHM}(P(\mu_{B,p}))$$

where $P(\mu_{B,p})$ denotes the empirical or kernel-estimated distribution of BP activity scores across patients.

In the implemented workflow, FWHM was estimated using a KDE-based procedure when possible. If the activity distribution was too sparse or unsuitable for stable KDE estimation, fallback procedures were used to avoid unstable width estimates.

Thus, each BP observable was represented as a point in (μ, W)space:

$$B \to (\mu_B, W_B)$$

where:

- μ_B represents observable activity center,
- W_B represents observable dispersion,
- W_B is interpreted as an operational proxy for uncertainty or state-cloud thickness.

This is the core D-PHY I representation.

S1.5 Observable coherence

To quantify whether a BP observable had strong activity relative to its uncertainty width, an observable coherence score was computed:

$$C_B = \frac{|\mu_B|}{W_B + \delta}$$

where δ is a small positive constant used to avoid division by zero.

Higher C_B indicates stronger activity concentration relative to dispersion. Lower C_B indicates either weak activity, large uncertainty width, or both.

Coherence was used in two ways:

1. to summarize uncertainty organization across BP observables;
2. to rank BP observables for scale-dependent analysis.

S1.6 Patient-level uncertainty-aware D-PHY score

To evaluate whether uncertainty structure improves clinical separation, patient-level uncertainty-aware scores were constructed.

For each patient, two summary quantities were computed across selected BP observables:

$$M_p = \mathrm{mean}_B(|\mu_{B,p}|)$$
$$U_p = \mathrm{mean}_B[\log(W_B + \delta)]$$

where:

- M_p represents the average BP activity magnitude for patient p,

- U_p represents the uncertainty burden derived from BP width values,
- B runs over the selected BP observables.

The patient-level uncertainty-aware score was then defined as:

$$D_{W\text{-}PHY,p} = M_p - \lambda U_p$$

where λ controls how strongly uncertainty width contributes to the final patient-level score.

This score does not mean that one patient has a literal “D value.” Rather, it is a patient-level observable used for survival stratification. The discriminability of this observable is then evaluated using survival analysis.

S1.7 λ-scan analysis

To evaluate the influence of uncertainty weighting, the analysis scanned the following values:

$$\lambda \in \{0, 0.25, 0.5, 0.75, 1.0, 1.5, 2.0\}$$

For each λ, patients were assigned a $D_{W\text{-}PHY,p}$-based score. Patients were then stratified into high- and low-score groups using median split, and survival separation was evaluated.

The optimal uncertainty weight was defined as:

$$\lambda^* = \arg\max_{\lambda} D_{\text{clinical}}(\lambda)$$

where:

$$D_{\text{clinical}}(\lambda) = -\log_{10}(p_{\text{log-rank}}(\lambda))$$

This λ-scan tested whether incorporating uncertainty width W improved clinical discriminability compared with activity-only scoring.

The special case:

$$\lambda = 0$$

corresponds to an activity-only baseline.

S1.8 Gene-set-level clinical discriminability

In addition to patient-level $D_{W\text{-}PHY}$ scoring, each BP gene set was evaluated individually.

For each BP observable B, two separate clinical discriminability values were computed:

$$D_{\mu}(B)$$

and

$$D_{W}(B)$$

where:

- $D_{\mu}(B)$ measures survival separation based on BP activity score,
- $D_{W}(B)$ measures survival separation based on uncertainty-related representation.

For each observable, patients were stratified using median split, Kaplan–Meier survival curves were generated, and log-rank p-values were transformed into discriminability scores.

The top discriminable BP observables were identified by ranking D_{μ} and D_{W}. Representative top observables were used to generate Kaplan–Meier survival plots and to support biological interpretation.

S1.9 Observation-scale analysis

Although the main focus of D-PHY I is uncertainty structure, the computational pipeline also included scale-dependent analysis as a companion module.

BP observables were ranked by coherence, and different numbers of top BP observables were selected:

$$K \in \{20, 50, 100, 200, 500\}$$

For each K, patient-level $D_{W\text{-}PHY,p}$ scores were recomputed, λ-scan analysis was repeated, and survival discriminability was evaluated.

The optimal observation scale was defined as:

$$K^{*} = \arg\max_{K} D(K)$$

This module is mainly used in AIDO-D-PHY II, but it was generated from the same computational framework and is therefore documented here for reproducibility.

S1.10 Structured-versus-random validation

To test whether BP observables exhibited non-random uncertainty and discriminability structure, size-matched random gene-set baselines were generated.

For each structured gene set, random gene sets of the same mapped size were sampled from the gene universe. Each random gene set was processed using the same workflow:

1. compute patient-level activity score;
2. estimate discriminability;
3. compare with the corresponding structured observable.

The formal random-test pipeline included three layers:

Layer 1: Structured versus random comparison

Structured BP observables were compared against size-matched random gene sets using summary statistics such as:

- structured mean D,
- structured maximum D,
- random mean D,
- random 95th percentile D,
- fraction of structured observables exceeding random 95th percentile.

Layer 2: High-D random dissection

High-discriminability random sets were extracted and examined for overlap with structured gene-set databases.

This step tested whether high-D random observables were purely stochastic or whether they partially captured hidden biological structure.

Layer 3: Cross-cancer recurrence

High-D random sets were compared across cancer cohorts to identify recurrent genes and recurrent gene-set signatures.

This step was used as an exploratory validation layer and was not treated as primary evidence in D-PHY I.

S1.11 Survival analysis and discriminability computation

For each patient-level observable, patients were divided into high- and low-score groups using median split.

Survival curves were estimated using Kaplan–Meier analysis, and group differences were evaluated using the log-rank test.

Clinical discriminability was defined as:

$$D_{\text{clinical}} = -\log_{10}(p_{\text{log-rank}})$$

where $p_{\text{log-rank}}$ is the p-value from the log-rank test.

This transformation converts survival separation into a positive continuous score, where larger values indicate stronger outcome separation.

S1.12 Output files and figure generation

The pipeline generated several categories of outputs:

BP uncertainty landscape outputs

- all gene-set summary tables
- (μ, W) statistics
- coherence values
- database-level summaries

These outputs support Figure 2.

Random validation outputs

- structured-versus-random summary
- high-D random set summaries
- random dissection tables

These outputs support Figure 3.

λ-scan outputs

- λ-dependent survival discriminability tables
- optimal λ summaries

These outputs support Figure 4.

K-scale outputs

- $D(K)$curves
- optimal K^*summaries
- K–λ interaction summaries

These outputs mainly support AIDO-D-PHY II.

Top BP outputs

- top $D_\mu$gene sets
- top $D_W$gene sets
- Kaplan–Meier plots for selected BP observables

These outputs support Figure 6 and supplementary biological interpretation.

S1.13 Summary of the analytical logic

The complete workflow can be summarized as:

Gene expression → BP activity → (μ, W) → $D_{W\text{-}PHY}$ → survival discriminability

The key methodological distinction of this study is that BP observables were not evaluated only by their activity values. Instead, each observable was evaluated jointly by activity magnitude, uncertainty width, coherence, and clinical discriminability.

This framework operationalizes the central hypothesis of AIDO-D-PHY I:

Transcriptomic biological-process observables are structured uncertainty-bearing projections of partially observable cancer systems, and their usefulness depends on both activity and uncertainty structure.

Supplementary Methods S2 | TCGA transcriptomic cohort preparation and preprocessing

This section describes the complete preprocessing workflow used to construct the transcriptomic observation matrices analyzed in the uncertainty-aware discriminability framework. The workflow was designed to maximize reproducibility and maintain consistent preprocessing across all cancer cohorts and pathway databases. The preprocessing pipeline corresponds to the formal implementation used in the AIDO-D random-test and D-PHY analysis framework.

S2.1 TCGA cohort acquisition

Transcriptomic and clinical datasets were obtained from the UCSC Xena platform using TCGA cancer cohorts. The datasets consisted primarily of bulk tumor transcriptomic profiles together with matched survival annotations.

The following cancer cohorts were included in the formal D-PHY analysis:

BLCA

BRCA

COAD

GBM

HNSC

KIRC

LAML

LIHC

LUAD

LUSC

OV

PAAD

PRAD

SARC

SKCM

STAD

THCA

UCEC

The formal pipeline used standardized cancer-folder mappings for automated processing across all cohorts.

Each cancer cohort was processed independently throughout all preprocessing, pathway scoring, uncertainty estimation, and discriminability analyses.

No pooled normalization across cancers was performed.

This design was used to preserve cancer-specific transcriptomic structure and avoid artificial compression caused by cross-cancer normalization.

S2.2 Expression matrix loading

For each cancer cohort, transcriptomic expression matrices were loaded from UCSC Xena cohort folders.

The preprocessing pipeline searched for expression files using the following priority order:

GE.tsv

HiSeqV2.tsv

expression.tsv

The formal implementation is shown below:

The first column of the expression matrix was interpreted as the gene identifier column, while remaining columns corresponded to patient samples.

The matrix was converted into the structure:

$$X \in \mathbb{R}^{G \times N}$$

where:

- G denotes the number of genes,
- N denotes the number of patients.

S2.3 Gene symbol normalization

Gene identifiers were standardized before downstream analysis.

The preprocessing workflow applied the following operations:

1. removal of transcript suffixes after the "|" delimiter;
2. trimming of whitespace characters;
3. conversion to uppercase gene symbols.

The formal implementation was:

For example:

TP53|7157 → TP53

egfr → EGFR

After normalization, duplicated genes were merged using mean aggregation:

$$x_g = \frac{1}{m} \sum_{i=1}^{m} x_{g,i}$$

where:

- $x_{g,i}$ represents duplicated entries for gene g,
- m is the number of duplicated rows.

The merged matrix therefore contained unique gene symbols only.

The formal duplicate-merging procedure was implemented using:

```
df = df.groupby(df.index).mean()
```

S2.4 Patient identifier standardization

TCGA sample identifiers were standardized using the first 12 characters of the barcode.

The preprocessing pipeline converted:

TCGA-XX-YYYY-01A...

into:

TCGA-XX-YYYY

This was implemented using:

```
return str(x).replace(".", "-")[:12]
```

The purpose of this procedure was to ensure consistent matching between transcriptomic and survival tables across TCGA file formats.

After standardization:

- duplicated sample identifiers were removed,
- only unique patient identifiers were retained.

S2.5 Survival-table loading and parsing

Clinical survival annotations were automatically detected from cohort folders.

The preprocessing workflow searched for files containing:

survival.tsv

If unavailable, fallback files were used:

Phenotype.tsv

phenotype.tsv

clinical.tsv

The pipeline automatically identified survival-time and event-status columns using prioritized candidate lists.

Time columns included:

OS.time

OS_Time

OS_Time_nature2012

OS.time.days

OS_days

time

days_to_death

DSS.time

PFI.time

Event columns included:

OS

OS_event

OS_event_nature2012

event

status

death

DSS

PFI

If standard labels were unavailable, fallback detection searched for columns containing keywords such as:

time

days

event

status

death

This automated detection allowed the same preprocessing pipeline to operate across heterogeneous TCGA clinical formats.

S2.6 Survival data cleaning

After survival columns were identified, the following filtering steps were applied:

1. remove samples with missing survival time;
2. remove samples with missing event labels;

3. remove samples with non-positive survival time;
4. remove duplicated patient identifiers.

The processed survival table was represented as:

$$S = \{(p_i, t_i, e_i)\}_{i=1}^{N}$$

where:

- p_i is patient ID,
- t_i is survival time,
- e_i is event status.

The pipeline also converted certain TCGA encodings:

1 = censored

2 = event

into binary form:

0 = censored

1 = event

using the transformation:

S2.7 Sample intersection between expression and survival tables

Only patients present in both the transcriptomic matrix and survival table were retained.

The common sample set was computed as:

$$P_{\text{common}} = P_{\text{expression}} \cap P_{\text{survival}}$$

The expression matrix and survival table were both restricted to this shared patient set.

This procedure ensured exact correspondence between transcriptomic observations and survival endpoints during downstream discriminability analysis.

The formal implementation was:

S2.8 Z-score normalization of transcriptomic expression

After sample matching, gene-expression values were standardized independently within each cancer cohort.

For gene g, z-score normalization was computed as:

$$z_{g,p} = \frac{x_{g,p} - \bar{x}_g}{s_g}$$

where:

- $x_{g,p}$ is the expression value of gene g in patient p,
- $\bar{x}_g$ is the mean expression of gene g,
- s_g is the standard deviation of gene g.

The resulting matrix:

$$Z \in \mathbb{R}^{G \times N}$$

was used for all downstream BP scoring analyses.

The formal implementation is shown below:

Genes with zero variance were handled using NaN replacement and zero filling to avoid instability during downstream averaging.

This normalization ensured that pathway scores reflected relative transcriptomic activity rather than absolute expression scale differences between genes.

S2.9 Quality-control thresholds

Several minimum-quality thresholds were enforced before survival analysis.

A cohort or pathway observation was excluded if:

Sample number too small

$$N < 30$$

Event number too small

$$E < 5$$

Median split produced unstable groups

Either group after median split contained:

$$n < 10$$

samples.

The formal implementation is shown in the survival-analysis routine.
These constraints were introduced to reduce unstable log-rank statistics caused by extremely small patient groups or sparse event counts.

S2.10 Biological interpretation of transcriptomic preprocessing

Within the proposed D-PHY framework, transcriptomic measurements were not interpreted as direct measurements of latent tumor states.
Instead, the normalized expression matrix:

$$Z$$

was interpreted as a partially observable projection generated through:

$$O = h(S) + \epsilon$$

where:

- S represents latent biological organization,
- $h(\cdot)$ represents the effective observation mapping,
- ϵ represents unresolved variability and measurement uncertainty.

This interpretation motivated the later introduction of pathway-level uncertainty width W and uncertainty-aware discriminability analysis in the main manuscript.

S2.11 Output objects generated from preprocessing

The preprocessing stage generated the following reusable objects for downstream analyses:

Expression objects

Expression matrix

Z-score normalized matrix

Gene universe

Clinical objects

Survival table

Event counts

Matched patient list

QC summaries

Sample counts

Event counts

Gene counts

Matched sample statistics

These outputs served as the direct input for:

1. pathway scoring;
2. uncertainty-width estimation;
3. uncertainty-aware discriminability analysis;
4. structured-versus-random validation.

Supplementary Methods S3 | Pathway database construction and pathway-level observable generation

This section describes the complete workflow used to construct pathway-level transcriptomic observables from normalized TCGA expression matrices. The objective of this stage was to transform high-dimensional gene-expression observations into biologically organized process-level observables suitable for uncertainty-width and discriminability analysis under partial observability.

The complete implementation corresponds to the formal AIDO-D Plan B and D-PHY processing pipelines.

S3.1 Conceptual rationale for pathway-level observables

Bulk transcriptomic measurements contain extremely high-dimensional and heterogeneous information arising from:

- tumor-cell populations,
- stromal contributions,
- immune infiltration,
- clonal variability,
- pathway cross-talk,
- unresolved regulatory interactions.

Direct interpretation at the individual-gene level is therefore highly unstable under partial observability.

To reduce dimensional instability while preserving biologically meaningful structure, transcriptomic observations were projected into pathway-level observables using curated biological-process gene sets.

Within the D-PHY framework, each pathway-level observable was interpreted as an effective projected representation:

$$O_B = h_B(S) + \epsilon_B$$

where:

- S denotes latent biological organization,
- h_B denotes the pathway-specific observation mapping,
- O_B denotes the pathway-level observable,
- ϵ_B represents unresolved variability and projection uncertainty.

This representation follows the partial-observability formulation described in the main manuscript.

S3.2 Pathway databases

Five curated biological pathway databases were used throughout the structured-versus-random and D-PHY analyses.

The formal database configuration was:

Hallmark

GO_BP

Reactome

KEGG_Medicus

WikiPathways

The corresponding GMT files were:

h.all.v2026.1.Hs.symbols.gmt

c5.go.bp.v2026.1.Hs.symbols.gmt

c2.cp.reactome.v2026.1.Hs.symbols.gmt

c2.cp.kegg_medicus.v2026.1.Hs.symbols.gmt

c2.cp.wikipathways.v2026.1.Hs.symbols.gmt

The pathway collections were obtained from MSigDB/GSEA-compatible repositories. Hallmark gene sets were originally described by Liberzon et al. [1], whereas GO:BP, Reactome, KEGG, and WikiPathways collections represent broader biological-process and signaling-network annotations [2–5].

S3.3 GMT parsing and gene-set loading

All pathway databases were loaded from GMT files.

Each GMT entry followed the structure:

GeneSetName<TAB>Description<TAB>Gene1<TAB>Gene2...

The parsing procedure extracted:

1. pathway name;
2. associated gene members.

The formal implementation was:

```
name = parts[0]
genes = sorted(set(
    standardize_gene_symbol(g)
    for g in parts[2:]
    if g.strip()
))
```

Gene symbols were standardized using the same normalization procedure described in Supplementary Methods S2.

Each pathway database was therefore represented as:

$$\mathcal{B} = \{B_1, B_2, \ldots, B_M\}$$

where:

- M is the number of pathway observables,
- B_i denotes the gene set associated with pathway i.

S3.4 Gene-set filtering and mapping

For each cancer cohort, pathway genes were intersected with the available transcriptomic gene universe:

$$B_i^{\text{mapped}} = B_i \cap G_{\text{expr}}$$

where:

- G_{expr} denotes genes present in the normalized expression matrix.

The formal implementation was:

```
mapped = sorted(
    set(genes).intersection(genes_available)
)
```

Only pathways satisfying the minimum mapped-gene threshold were retained.

The formal threshold was:

$$| B_i^{\text{mapped}} | \geq 5$$

defined by:

```
MIN_MAPPED_GENES = 5
```

For structured-versus-random analyses, an additional filtering range was used:

$$10 \leq | B_i^{\text{mapped}} | \leq 300$$

to reduce instability associated with extremely small or excessively broad pathways.

The rationale for this filtering step was:

- extremely small pathways produce unstable estimates,
- extremely large pathways dilute pathway specificity through excessive averaging.

S3.5 Pathway-level activity calculation

For each retained pathway B_i, pathway activity was computed using the mean z-scored expression across mapped genes.

For patient p:

$$\mu_{i,p} = \frac{1}{| B_i |} \sum_{g \in B_i} z_{g,p}$$

where:

- $z_{g,p}$ is normalized expression,
- $\mu_{i,p}$ is pathway activity center.

The formal implementation was:

```
mu = mat.mean(axis=0)
```

This generated the pathway-activity matrix:

$$\boldsymbol{\mu} \in \mathbb{R}^{M\times N}$$

where:

- M is the number of pathways,
- N is the number of patients.

Within the D-PHY framework, μ was interpreted as the center of the observable state cloud.

S3.6 Pathway-level uncertainty-width estimation

To quantify uncertainty structure, pathway-level observable width W was computed for each patient-pathway pair.

For pathway B_i and patient p, the gene-expression distribution:

$$\{z_{g,p}: g \in B_i\}$$

was treated as an observable cloud.

The pathway width:

$$W_{i,p}$$

was estimated using empirical full width at half maximum (FWHM).

The primary implementation used kernel-density estimation (KDE)-based FWHM estimation whenever possible.

The KDE procedure was:

1. estimate pathway-expression density;
2. identify half-maximum density level;
3. compute the width of the density region above half maximum.

Formally:

$$W_{i,p} = x_{\text{right}} - x_{\text{left}}$$

where:

- x_{left} and x_{right} denote half-maximum boundaries.

If KDE estimation failed or pathway size was insufficient, a robust IQR-based approximation was used:

$$W \approx 2.355 \times \frac{\text{IQR}}{1.349}$$

The fallback implementation was:

Within the D-PHY framework:

- μrepresents cloud center,
- Wrepresents state-cloud thickness.

This interpretation forms the basis of uncertainty-aware discriminability.

S3.7 Construction of pathway-level observable matrices

For each pathway database and cancer cohort, two matrices were generated:

Activity matrix

$$\boldsymbol{\mu} = [\mu_{i,p}]$$

Width matrix

$$\mathbf{W} = [W_{i,p}]$$

The formal implementation was:

mu_matrix = pd.DataFrame(mu_records)

w_matrix = pd.DataFrame(w_records)

Each row represented one pathway observable.

Each column represented one patient.

These matrices served as the direct input for:

1. coherence analysis;
2. D_W-PHY calculation;
3. scale-dependent discriminability analysis;
4. structured-versus-random comparisons.

S3.8 Pathway-level summary statistics

For each pathway, additional summary statistics were computed.

The following quantities were recorded:

mu_mean_abs

mu_mean

W_mean

W_median

W_sd

coherence_mean

These correspond to:

Mean absolute pathway activity

$$\langle| \mu_i |\rangle$$

Mean uncertainty width

$$\langle W_i \rangle$$

Mean coherence

$$C_i = \langle\frac{| \mu_i |}{W_i + \delta}\rangle$$

where:

- $\delta = 10^{-6}$

was introduced for numerical stability.

These summary quantities were later used for:

- pathway ranking,
- uncertainty-landscape visualization,
- K-scale observation analysis,
- structured-versus-random comparisons.

S3.9 Controlled pathway subsets

To reduce pathway-size bias, controlled analyses were performed using pathways satisfying:

$$10 \leq | B_i | \leq 300$$

The controlled pathway subset was generated using:

```
controlled = summary_all[
    (summary_all["n_genes_mapped"] >= CONTROL_MIN_GENES) &
    (summary_all["n_genes_mapped"] <= CONTROL_MAX_GENES)
```

]

This restriction was introduced because:

- very small pathways exhibit unstable width estimation,
- very large pathways excessively smooth transcriptomic structure.

Controlled analyses therefore focused on biologically interpretable intermediate-scale observables.

S3.10 Biological interpretation of pathway observables

Within the D-PHY framework, pathway observables were interpreted as uncertainty-bearing projections rather than direct biological-state measurements. For pathway B_i:

$$(\mu_i, W_i)$$

defines the effective observable state cloud.

Conceptually:

Pathway center

$$\mu_i$$

represents effective process activity.

Pathway width

$$W_i$$

represents uncertainty thickness arising from:

- latent heterogeneity,
- projection compression,
- pathway overlap,
- unresolved biological variability,
- microenvironmental mixing.

This interpretation follows the state-cloud representation described in the main manuscript.

S3.11 Output objects generated from pathway construction

The pathway-processing stage generated:

Patient-level matrices

μ matrix

W matrix

D_W-PHY score matrix

Pathway-level summaries

Mean μ

Mean W

Coherence

Mapped gene count

Controlled pathway subsets

10–300 gene pathways

Top coherent pathways

Top narrow-width pathways

These outputs were subsequently used in:

1. uncertainty-aware discriminability analysis;
2. scale-dependent D-PHY analysis;
3. cross-cancer comparisons;
4. structured-versus-random validation experiments.

References

1. The Molecular Signatures Database hallmark gene set collection
2. Gene set enrichment analysis: a knowledge-based approach for interpreting genome-wide expression profiles
3. Gene Ontology Consortium
4. Reactome
5. WikiPathways

Supplementary Methods S4 | Observable coherence, uncertainty-aware discriminability, and λ-scan analysis

This section describes the complete computational workflow used to construct observable coherence measures, patient-level uncertainty-aware scores, clinical discriminability metrics, and λ-dependent uncertainty-penalty analyses within the D-PHY framework.

The workflow described here corresponds directly to the formal implementation used in the D-PHY Plan B and K-scale analysis pipelines.

S4.1 Conceptual rationale

Within the proposed D-PHY framework, pathway-level transcriptomic observables are interpreted as uncertainty-bearing projections of latent biological organization rather than direct measurements of tumor state.

Each pathway observable is represented by:

$$(\mu, W)$$

where:

- μ denotes observable activity center,
- W denotes uncertainty thickness or state-cloud width.

Under this representation, observable quality is not determined solely by activity magnitude.

Instead, clinically useful observables are expected to exhibit:

1. sufficiently strong observable activity;
2. sufficiently coherent uncertainty organization;
3. stable clinical outcome separation.

This motivated the introduction of:

- pathway coherence,
- uncertainty-aware discriminability,
- λ-dependent uncertainty penalization.

The theoretical rationale is discussed in the main manuscript.

S4.2 Pathway-level coherence calculation

For each pathway B_i, coherence was defined as the ratio between activity magnitude and uncertainty width.

The pathway-level coherence score was computed as:

$$C_i = \frac{\mid \mu_i \mid}{W_i + \delta}$$

where:

- μ_i denotes pathway activity center,
- W_i denotes pathway uncertainty width,
- δ is a small numerical stabilization constant.

The formal implementation used:

$$\delta = 10^{-6}$$

defined as:

DELTA = 1e-6

The pathway-level mean coherence across patients was computed as:

$$\langle C_i \rangle = \frac{1}{N} \sum_{p=1}^{N} \frac{\mid \mu_{i,p} \mid}{W_{i,p} + \delta}$$

where:

- N denotes patient number.

The formal implementation was:

coherence = np.abs(mu) / (W + DELTA)

S4.3 Biological interpretation of coherence

Within the D-PHY framework:

High coherence

corresponds to:

- strong observable activity,
- narrow uncertainty thickness,
- concentrated observable organization.

Low coherence

corresponds to:

- weak activity,
- broad uncertainty clouds,
- high projection ambiguity.

Importantly, coherence was not interpreted merely as signal-to-noise ratio.

Instead, coherence represents the balance between:

- observable magnitude,
- latent uncertainty organization,
- projection compression.

Thus, coherence measures how effectively a pathway preserves informative structure under partial observability.

This interpretation follows the state-cloud representation proposed in the main manuscript.

S4.4 Construction of patient-level magnitude and uncertainty terms

For each patient p, pathway-level observables were aggregated into two summary quantities.

S4.4.1 Mean activity magnitude

The average absolute pathway activity was computed as:

$$M_p = \frac{1}{K}\sum_{i=1}^{K} |\mu_{i,p}|$$

where:

- K denotes the number of selected pathways,
- $\mu_{i,p}$ denotes pathway activity.

The formal implementation was:

```
M = np.mean(np.abs(mu_vals))
```

M_p represents the average observable activity magnitude across selected pathways.

S4.4.2 Mean uncertainty burden

The uncertainty contribution was computed as:

$$U_p = \frac{1}{K}\sum_{i=1}^{K} \log(W_{i,p} + \delta)$$

where:

- $W_{i,p}$ denotes pathway width,
- logarithmic transformation reduces dominance from extreme width values.

The formal implementation was:

U = np.mean(np.log(W_vals + DELTA))

Within the D-PHY framework:

- M_p measures observable magnitude,
- U_p measures uncertainty burden.

S4.5 Construction of patient-level uncertainty-aware score

For each patient, the uncertainty-aware D-PHY score was computed as:

$$D_{W\text{-}PHY,p} = M_p - \lambda U_p$$

where:

- M_p represents average pathway activity magnitude,
- U_p represents uncertainty contribution,
- λ controls uncertainty penalization strength.

The formal implementation was:

score = M - lam * U

Importantly, this quantity is not itself the final discriminability value.

Instead:

$$D_{W\text{-}PHY,p}$$

is a patient-level observable score used for downstream survival stratification. Clinical discriminability is then evaluated from survival separation generated by this score.

S4.6 λ-scan analysis

To evaluate the contribution of uncertainty structure, multiple uncertainty-penalty strengths were examined.

The formal λ grid was:

LAMBDA_GRID = [0, 0.25, 0.5, 0.75, 1.0, 1.5, 2.0]

Correspondingly:

$$\lambda \in \{0, 025, 0.50, 0.75, 1.0, 1.5, 2.0\}$$

For each λ:

1. patient-level D-PHY scores were recomputed;
2. survival stratification was performed;
3. clinical discriminability was evaluated.

The λ-scan therefore tested whether uncertainty-width incorporation improved observable quality relative to activity-only scoring.

S4.7 Interpretation of λ

Within the proposed framework:

λ = 0

corresponds to:

- activity-only scoring,
- no uncertainty penalty.

This acts as the baseline model.

λ > 0

introduces uncertainty penalization.

Pathways with broader uncertainty width contribute less strongly to the final patient-level score.

Large λ

strongly suppresses broad uncertainty clouds.

Excessively large λ values may therefore:

- over-penalize biologically relevant pathways,
- reduce effective discriminability.

This motivates the existence of cancer-specific optimal uncertainty penalties.

S4.8 Clinical survival stratification

For each λ value, patients were stratified into:

- high-score group,

- low-score group,

using median split:

$$\text{Group}_{high} = \{p: D_{W\text{-}PHY,p} > \text{median}\}$$
$$\text{Group}_{low} = \{p: D_{W\text{-}PHY,p} \leq \text{median}\}$$

The formal implementation was:

```
median_score = np.median(scores)
high = scores > median_score
low = scores <= median_score
```

Median split was selected because:

- it provides balanced survival groups,
- reduces instability caused by highly unequal partition sizes,
- enables consistent cross-cancer comparison.

S4.9 Kaplan–Meier survival analysis

For each stratification:

1. Kaplan–Meier survival curves were estimated;
2. log-rank tests were performed;
3. survival separation was converted into discriminability values.

Kaplan–Meier estimation followed the standard survival formulation:

$$\hat{S}(t) = \prod_{t_i \leq t} \left(1 - \frac{d_i}{n_i}\right)$$

where:

- d_i is the number of events,
- n_i is the number at risk.

The formal implementation used the lifelines Python library.

S4.10 Clinical discriminability calculation

Clinical discriminability was defined as:

$$D_{\text{clinical}} = -\log_{10}(p_{\text{log-rank}})$$

where:

- $p_{\text{log-rank}}$is the log-rank test p-value.

The formal implementation was:

D = -np.log10(max(p_value, 1e-300))

This transformation converts smaller p-values into larger positive discriminability values.

Thus:

Larger D

indicates:

- stronger survival separation,
- higher clinical discriminability.

S4.11 Optimal λ identification

For each cancer cohort, the optimal uncertainty penalty was defined as:

$$\lambda^* = \arg \max_{\lambda} D_{\text{clinical}}(\lambda)$$

The formal implementation was:

best_idx = np.argmax(D_values)

best_lambda = lambdas[best_idx]

The resulting:

$$\lambda^*$$

represents the uncertainty weighting producing maximal survival discriminability.

S4.12 Cancer-specific uncertainty landscapes

The λ-scan generated cancer-specific uncertainty-discriminability landscapes:

$$D(\lambda)$$

These landscapes characterize how discriminability changes as uncertainty structure is progressively incorporated into pathway scoring.

Different cancers exhibited:

- distinct optimal λ values,
- distinct uncertainty sensitivity,
- distinct D(λ) curve shapes.

This observation supports the interpretation that cancer systems exhibit heterogeneous observable uncertainty organization.

S4.13 Stability and numerical constraints

Several numerical safeguards were implemented during uncertainty-aware scoring.

S4.13.1 Width stabilization

$$W + \delta$$

was used to avoid logarithmic instability.

S4.13.2 Extreme p-value stabilization

Minimum p-value clipping:

$$p \geq 10^{-300}$$

prevented infinite discriminability values.

S4.13.3 Group-size constraints

Survival analysis was skipped if:

$$n_{group} < 10$$

or:

$$E < 5$$

where:

- n_{group} is group size,
- E is event count.

These constraints reduced unstable survival estimates.

S4.14 Output objects generated from uncertainty-aware analysis

The D-PHY scoring workflow generated:

Patient-level outputs

D_W-PHY scores

M values

U values

λ-specific scores

Survival outputs

Kaplan–Meier curves

Log-rank p-values

Clinical D values

Cancer-level outputs

Optimal λ

D(λ) curves

Maximum D

These outputs directly supported:

- Figure 2,
- Figure 4,
- uncertainty-aware pathway ranking,
- scale-dependent discriminability analysis.

S4.15 Biological interpretation of uncertainty-aware discriminability

Within the D-PHY framework:

$$D_{W\text{-}PHY}$$

does not merely represent statistical survival separation.

Instead, it represents an operational measure of how effectively biological observables preserve clinically relevant structure under partial observability.

In this interpretation:

High discriminability

emerges from:

- strong observable organization,
- coherent uncertainty structure,
- reduced projection ambiguity.

Low discriminability

emerges from:

- diffuse uncertainty clouds,
- broad projection overlap,
- weak observable organization.

Thus, discriminability is interpreted as an emergent property of observable geometry rather than merely a statistical endpoint.

Supplementary Methods S5 | Scale-dependent observation analysis and K-scale discriminability framework

This section describes the complete workflow used to evaluate scale-dependent observable organization within the D-PHY framework. The objective of this analysis was to determine whether uncertainty-aware discriminability changes systematically with observation scale and whether partially observable cancer systems exhibit optimal transcriptomic observation scales.

The complete implementation corresponds to the formal K-scale analysis pipeline used in D-PHY II and integrated D-PHY analyses.

S5.1 Conceptual rationale for observation-scale analysis

Modern transcriptomic analysis often assumes that increasing the number of molecular observables improves biological inference.

However, under partial observability, additional observables may simultaneously introduce:

- redundancy,
- projection overlap,
- uncertainty accumulation,
- latent-state compression.

Consequently, observational complexity does not necessarily increase informational quality monotonically.

Within the D-PHY framework, transcriptomic observables are interpreted as projections:

$$O = h(S) + \epsilon$$

where increasing observable dimensionality may either:

1. improve latent-state coverage, or
2. dilute informative structure through projection mixing.

This motivated the introduction of scale-dependent discriminability analysis.

The central hypothesis was:

There exists a nontrivial optimal observation scale under partial observability.

The theoretical rationale is described in the main manuscript.

S5.2 Definition of observation scale K

Observation scale was defined as the number of selected pathway observables used to construct the patient-level uncertainty-aware score.

For each cancer cohort:

$$K = \text{number of selected BP observables}$$

The formal K-scale grid was:

K_VALUES = [20, 50, 100, 200, 500]

Correspondingly:

$$K \in \{20, 50, 100, 200, 500\}$$

These values were selected to span:

- small-scale observation,
- intermediate-scale observation,
- large-scale observation.

S5.3 Selection of pathway observables for K-scale analysis

Before K-scale construction, pathway observables were ranked according to coherence.

For pathway B_i:

$$C_i = \frac{|\mu_i|}{W_i + \delta}$$

where:

- μ_i is pathway activity center,
- W_i is uncertainty width.

The formal implementation ranked pathways using:

summary.sort_values("coherence_mean", ascending=False)

The top K pathways were then selected:

$$\mathcal{B}_K = \{B_1, B_2, \dots, B_K\}$$

where pathways were ordered by decreasing coherence.

Thus:

- small K corresponds to highly selective observation,
- large K corresponds to broader observational coverage.

S5.4 Construction of K-dependent patient scores

For each scale K, patient-level magnitude and uncertainty terms were recomputed using only the selected pathway subset:

$$\mathcal{B}_K$$

The K-dependent activity term was:

$$M_p^{(K)} = \frac{1}{K}\sum_{i=1}^{K} | \mu_{i,p} |$$

The K-dependent uncertainty term was:

$$U_p^{(K)} = \frac{1}{K}\sum_{i=1}^{K} \log (W_{i,p} + \delta)$$

The corresponding K-dependent uncertainty-aware score was:

$$D_{W\text{-}PHY,p}^{(K)} = M_p^{(K)} - \lambda U_p^{(K)}$$

The formal implementation was:

score = M - lam * U

where the pathway subset depended explicitly on K.

S5.5 λ-scan within each observation scale

For each observation scale K, a complete λ-scan analysis was repeated independently.

Thus:

$$D(K,\lambda)$$

was computed over the full parameter grid:

$$K \in \{20,50,100,200,500\}$$

$$\lambda \in \{0,0.25,0.5,0.75,1.0,1.5,2.0\}$$

The resulting two-dimensional landscape represents the interaction between:

- observational coverage,
- uncertainty penalization.

The formal nested implementation was:

```
for K in K_VALUES:
    for lam in LAMBDA_GRID:
```

S5.6 Survival stratification at each observation scale

For each (K, λ) pair:

1. patient-level scores were computed;
2. patients were divided into high/low groups using median split;
3. Kaplan–Meier survival analysis was performed;
4. log-rank p-values were computed.

Median split was performed as:

$$\text{Group}_{high} = \{p: D_{W\text{-}PHY,p}^{(K)} > \text{median}\}$$

$$\text{Group}_{low} = \{p: D_{W\text{-}PHY,p}^{(K)} \leq \text{median}\}$$

The resulting survival separation was converted into clinical discriminability:

$$D_{\text{clinical}}(K,\lambda) = -\log_{10}(p_{\text{log-rank}})$$

S5.7 Determination of optimal observation scale

For each cancer cohort, the maximal discriminability at scale K was defined as:

$$D(K) = \max_{\lambda} D_{\text{clinical}}(K, \lambda)$$

The optimal observation scale was then defined as:

$$K^* = \arg\max_{K} D(K)$$

The formal implementation was:

```
best_idx = np.argmax(D_values)
best_K = K_values[best_idx]
```

Thus:

- K^* denotes the observation scale producing maximal clinical discriminability.

S5.8 Interpretation of observation scales

Within the D-PHY framework:

Small observation scales

Small Kvalues correspond to highly selective observations.

Advantages:

- reduced projection overlap,
- reduced uncertainty accumulation,
- high pathway specificity.

Limitations:

- insufficient latent-system coverage,
- omission of relevant biological structure.

Large observation scales

Large Kvalues correspond to broad observational coverage.

Advantages:

- wider biological-process inclusion,
- increased latent-state sampling.

Limitations:

- redundancy,
- pathway overlap,
- uncertainty accumulation,
- projection dilution.

Intermediate observation scales

Intermediate K values may balance:

- informational coverage,
- uncertainty compression.

This balance motivates the emergence of optimal observation scales.

S5.9 Construction of D(K) curves

For each cancer cohort, the scale-dependent discriminability curve:

$$D(K)$$

was generated by plotting maximal discriminability against observation scale.

These curves characterize how observable quality changes with transcriptomic observation complexity.

Different cancers exhibited distinct curve morphologies, including:

- monotonic increase,
- intermediate peaks,
- flat responses,
- unstable responses.

These curve structures were later used for scale-regime interpretation.

S5.10 Scale-regime classification

To summarize scale-dependent behavior across cancers, observational regimes were classified heuristically according to the shape of $D(K)$.

The main qualitative regimes included:

Small-scale dominant

$$K^* \leq 50$$

Interpretation:

- informative structure concentrated in highly coherent pathways,
- broader pathway inclusion introduces dilution.

Intermediate-optimal

$$50 < K^* < 500$$

Interpretation:

- balance between informational coverage and uncertainty accumulation.

Large-scale dominant

$$K^* \geq 500$$

Interpretation:

- broader pathway integration improves observable organization.

Flat / weak scale dependence

$$D(K) \approx \text{constant}$$

Interpretation:

- observation scale weakly influences discriminability.

Importantly, these classifications were exploratory and descriptive rather than formal statistical categories.

S5.11 K–λ interaction analysis

To investigate whether observation scale interacts with uncertainty penalization, the relationship between:

$$K^*$$

and:

$$\lambda^*$$

was analyzed across cancers.

For each cancer cohort:

- optimal scale,
- optimal λ,
- maximal discriminability

were jointly recorded.

The resulting K–λ relationship was summarized using:

- scatter plots,
- heatmaps,
- correlation analysis.

This analysis examined whether broader observational scales require stronger uncertainty regularization.

S5.12 Heatmap construction

Cross-cancer K-scale heatmaps were generated using:

$$D(K)$$

or:

$$D(K,\lambda)$$

values.

Rows corresponded to cancer cohorts.

Columns corresponded to observation scales.

Heatmap values represented maximal discriminability.

These heatmaps visualize the global landscape of scale-dependent observable organization.

S5.13 Numerical and stability constraints

Several constraints were implemented during K-scale analysis.

Minimum patient number

$$N \geq 30$$

Minimum event count

$$E \geq 5$$

Minimum survival-group size

$$n_{\text{group}} \geq 10$$

Width stabilization

$$W + \delta$$

with:

$$\delta = 10^{-6}$$

Minimum p-value clipping

$$p \geq 10^{-300}$$

These constraints reduced unstable discriminability estimates caused by sparse survival structure.

S5.14 Output objects generated from K-scale analysis

The K-scale workflow generated:

Scale-dependent outputs

D(K) curves

D(K,λ) landscapes

Optimal K*

Optimal λ*

Cross-cancer summaries

Scale-regime classification

K–λ relationship

Heatmap summaries

Visualization outputs

Small-multiple D(K) plots

Scale heatmaps

K* bar plots

K–λ interaction plots

These outputs directly supported:

- D-PHY II Figure 2,
- Figure 3,
- Figure 4,
- scale-regime interpretation.

S5.15 Biological interpretation of scale-dependent discriminability

Within the D-PHY framework:

$$D(K)$$

is interpreted as a measure of how effectively a given observational scale preserves clinically relevant structure under partial observability.

Thus:

Increasing K

does not necessarily improve information quality monotonically.

Instead, increasing observational complexity may simultaneously:

- increase informational coverage,
- increase uncertainty accumulation,
- increase projection overlap.

Consequently, observable quality emerges from the balance between:

1. latent-state coverage,
2. uncertainty compression,
3. observational redundancy.

This interpretation forms the conceptual basis of scale-dependent discriminability in partially observable cancer systems.

Supplementary Methods S6 | Structured-versus-random validation

framework and random observable analysis

This section describes the complete structured-versus-random validation workflow used to evaluate whether the uncertainty-aware organization observed in biological pathway observables exceeds the structure expected from random gene aggregation alone.

The analysis was designed to determine whether:

1. biologically curated pathways exhibit higher discriminability than random observables;
2. observable uncertainty organization can emerge partially from generic high-dimensional statistical structure;
3. biologically constrained pathways reshape the statistical observable landscape into more coherent uncertainty organization.

The formal implementation corresponds to the AIDO-D Random Test V1.3 framework.

S6.1 Conceptual rationale

High-dimensional transcriptomic spaces intrinsically contain nontrivial statistical structure.

Therefore, biologically meaningful discriminability cannot be assumed solely because a pathway or gene set exhibits significant survival separation.

Previous studies demonstrated that random gene signatures may also generate significant outcome associations in cancer transcriptomics [1].

Consequently, the present study explicitly compared:

- structured pathway observables,
- size-matched random gene sets.

Within the D-PHY framework, the structured-versus-random analysis evaluates whether biological pathway organization preserves additional uncertainty-aware structure beyond generic high-dimensional statistical effects.

The random-validation workflow was organized into three layers:

1. structured-versus-random baseline comparison;
2. high-discriminability random-set dissection;
3. cross-cancer recurrence analysis.

S6.2 Random-test cancer cohorts and pathway databases

The structured-versus-random workflow was applied across the same cancer cohorts used in the D-PHY analysis.

The formal cohort list was:

BLCA

BRCA

COAD

GBM

HNSC

KIRC

LAML

LIHC

LUAD

LUSC

OV

PAAD

PRAD

SARC

SKCM

STAD

THCA

UCEC

The following pathway databases were included:

Hallmark

GO_BP

Reactome

KEGG_Medicus

WikiPathways

These databases span multiple levels of biological-process and signaling-network organization.

S6.3 Gene-universe construction

For each cancer cohort, a transcriptomic gene universe was constructed from the normalized expression matrix.

The available gene set was defined as:

$$G_{\text{expr}} = \{g_1, g_2, \dots, g_n\}$$

where genes satisfied preprocessing and normalization requirements described in Supplementary Methods S2.

The formal implementation was:

```
genes_available = set(expr_df.index)
```

Only genes present in the processed transcriptomic matrix were eligible for random sampling.

S6.4 Structured pathway filtering before random comparison

Before randomization, pathway observables were filtered using the same constraints applied in the main D-PHY analysis.

For pathway B_i:

$$B_i^{\text{mapped}} = B_i \cap G_{\text{expr}}$$

Only pathways satisfying:

$$10 \leq | B_i^{\text{mapped}} | \leq 300$$

were retained for formal random comparison.

The formal implementation was:

```
if mapped_size < CONTROL_MIN_GENES:
    continue
if mapped_size > CONTROL_MAX_GENES:
    continue
```

This restriction reduced:

- instability from very small pathways,
- excessive smoothing from extremely large pathways.

S6.5 Construction of size-matched random gene sets

For each structured pathway B_i, random gene sets were generated using size-matched sampling from the available transcriptomic gene universe.

For pathway size:

$$|B_i| = m$$

a random gene set:

$$R_i = \{r_1, r_2, \dots, r_m\}$$

was sampled uniformly from:

$$G_{\text{expr}}$$

subject to:

$$|R_i| = |B_i|$$

The formal implementation was:

random.sample(sorted(genes_available), mapped_size)

Thus, each structured pathway was compared against random observables with identical mapped gene counts.

S6.6 Number of random repetitions

For each structured pathway, multiple independent random gene sets were generated.

The formal implementation used:

$$N_{\text{random}} = 20$$

defined as:

N_RANDOM_PER_SET = 20

Thus:

- each structured pathway generated 20 size-matched random controls.

This repeated sampling allowed estimation of empirical random discriminability distributions.

S6.7 Processing of random gene sets

Each random gene set was processed using exactly the same workflow applied to structured pathways.

For each random set:

1. pathway activity μ was computed;
2. uncertainty width W was estimated;
3. coherence was calculated;
4. uncertainty-aware scoring was performed;
5. survival discriminability was evaluated.

Thus, random observables were analyzed under identical preprocessing, scoring, and survival-analysis procedures.

This ensured that differences between structured and random observables were not caused by pipeline asymmetry.

S6.8 Clinical discriminability of random observables

For each random gene set:

$$R_i$$

clinical discriminability was computed as:

$$D_{\text{random}} = -\log_{10}(p_{\text{log-rank}})$$

using the same Kaplan–Meier and log-rank workflow described previously.

The resulting random distributions were used to estimate:

- random mean D,
- random maximum D,
- random 95th percentile D.

S6.9 Layer 1 | Structured-versus-random comparison

The first validation layer compared global discriminability statistics between:

- structured pathway observables,
- size-matched random observables.

The following quantities were computed:

Structured mean discriminability

$$\langle D_{\text{structured}} \rangle$$

Structured maximum discriminability

$$D_{\text{structured}}^{\max}$$

Random mean discriminability

$$\langle D_{\text{random}} \rangle$$

Random 95th percentile

$$Q_{95}^{\text{random}}$$

Structured fraction exceeding random baseline

$$F_{95} = \frac{\#(D_{\text{structured}} > Q_{95}^{\text{random}})}{N_{\text{structured}}}$$

The formal implementation was:

These statistics quantify whether biologically organized pathways preserve discriminability beyond random expectation.

S6.10 Layer 2 | High-discriminability random-set dissection

The second validation layer investigated whether high-discriminability random gene sets are purely stochastic or partially biologically structured.

Random gene sets satisfying:

$$D_{\text{random}} \geq Q_{95}^{\text{random}}$$

were extracted as:

High-D random sets

The formal implementation was:

For each high-D random set:

1. gene composition was recorded;
2. pathway overlap was evaluated;
3. recurrent genes were counted.

This analysis tested whether latent biological axes remain partially embedded within random transcriptomic sampling.

S6.11 Layer 3 | Cross-cancer recurrence analysis

The third validation layer examined whether genes appearing in high-discriminability random sets recur across cancers.

For gene g:

$$R_g = \text{number of cancers where } g \text{ appears in high-D random sets}$$

The recurrence analysis therefore identified genes repeatedly associated with high random discriminability across independent cancer cohorts.

The formal implementation was:

The resulting recurrence distributions were used as exploratory indicators of:

- latent pan-cancer transcriptomic structure,
- hidden biological axes,
- recurrent uncertainty geometry.

Importantly, these recurrence analyses were treated as exploratory rather than definitive mechanistic evidence.

S6.12 Construction of random-baseline summary statistics

For each cancer and pathway database, the following summary outputs were generated:

Structured statistics

Structured mean D

Structured maximum D

Structured median D

Random statistics

Random mean D

Random maximum D

Random 95th percentile D

Comparative statistics

Fraction above random 95th percentile

Structured/random ratio

Delta D

These statistics formed the basis of:

- Figure 3,
- Figure 5,
- cross-cancer structured-versus-random comparisons.

S6.13 Statistical interpretation of random observables

Within the D-PHY framework, the presence of high-discriminability random observables does not imply that biological pathways are meaningless.

Instead, the results suggest that:

1. high-dimensional transcriptomic spaces possess intrinsic statistical structure;
2. biological pathways reshape this statistical structure into more coherent observable organization.

Thus:

Random observables

represent:

- baseline statistical geometry.

Structured pathways

represent:

- biologically constrained reshaping of observable geometry.

This interpretation follows the conceptual framework proposed in the main manuscript.

S6.14 Numerical constraints and reproducibility safeguards

Several safeguards were implemented during random analysis.

Fixed random repetitions

$$N_{\text{random}} = 20$$

Size matching

$$|R_i| = |B_i|$$

Controlled pathway size range

$$10 \leq |B_i| \leq 300$$

Shared preprocessing pipeline

Structured and random observables used:

- identical expression matrices,
- identical normalization,
- identical survival procedures,
- identical discriminability calculations.

Survival constraints

Analysis skipped if:

$$n_{\text{group}} < 10$$

or:

$$E < 5$$

These constraints reduced unstable random discriminability estimates.

S6.15 Output objects generated from random analysis

The random-test framework generated:

Structured-versus-random summaries

Mean D

Maximum D

Random 95th percentile

Fraction above random baseline

High-D random outputs

High-D random gene sets

Gene overlap tables

Enrichment summaries

Cross-cancer recurrence outputs

Gene recurrence tables

Cross-cancer recurrence counts

Visualization outputs

Structured vs random plots

Random-baseline histograms

Recurrence plots

These outputs directly supported:

- Figure 3,
- Figure 5,
- structured-versus-random validation analyses.

S6.16 Biological interpretation of structured-versus-random organization

The structured-versus-random framework supports a key D-PHY interpretation:

Biological observables emerge from the interaction between intrinsic statistical structure and biologically constrained pathway organization.

Within this view:

Random transcriptomic spaces

already possess nontrivial observable geometry.

Biological pathways

reshape this geometry into:

- narrower uncertainty organization,
- higher coherence,
- stronger discriminability.

Thus, pathway-level observable quality is interpreted as an emergent property of:

1. statistical structure,
2. biological organization,
3. uncertainty geometry under partial observability.

References

1. Most random gene expression signatures are significantly associated with breast cancer outcome

Supplementary Methods S7 | Statistical analysis, visualization workflow, and reproducibility framework

This section describes the statistical procedures, numerical safeguards, visualization workflows, and reproducibility framework used throughout the D-PHY analyses. The purpose of this section is to provide sufficient detail for complete computational reproduction of all reported results, figures, and summary statistics.

The procedures described here correspond directly to the formal implementation used in the D-PHY Plan B, K-scale, and Random-Test pipelines.

S7.1 Computational environment

All analyses were performed using Python-based computational pipelines.

The primary Python packages included:

numpy

pandas

scipy

matplotlib

seaborn

lifelines

scikit-learn

The main workflow components included:

1. transcriptomic preprocessing;
2. pathway-level observable generation;
3. uncertainty-width estimation;
4. coherence analysis;
5. uncertainty-aware discriminability analysis;
6. structured-versus-random validation;
7. K-scale observation analysis;
8. visualization and summary generation.

The entire workflow was executed using reproducible scripted pipelines rather than manual interactive processing.

S7.2 Numerical precision and floating-point safeguards

Several numerical stabilization procedures were implemented to prevent instability during large-scale transcriptomic processing.

S7.2.1 Width stabilization

To prevent division-by-zero and logarithmic instability, all width calculations used:

$$W + \delta$$

with:

$$\delta = 10^{-6}$$

defined as:

DELTA = 1e-6

This stabilization was applied consistently across:

- coherence calculations,
- uncertainty scoring,
- logarithmic transformations.

S7.2.2 Extreme p-value stabilization

Very small survival p-values were clipped using:

$$p \geq 10^{-300}$$

before discriminability transformation.

The formal implementation was:

D = -np.log10(max(p_value, 1e-300))

This prevented infinite discriminability values caused by floating-point underflow.

S7.2.3 NaN and infinite-value handling

During preprocessing and pathway scoring:

- NaN values,
- infinite values,
- zero-variance genes

were replaced using stable fallback procedures.

The preprocessing implementation included:

replace([np.inf, -np.inf], np.nan)

fillna(0)

This ensured stable downstream averaging and FWHM estimation.

S7.3 Survival-analysis procedures

All survival analyses were performed independently for each cancer cohort.

For each patient-level observable:

1. patients were stratified into high-score and low-score groups using median split;
2. Kaplan–Meier survival curves were estimated;
3. log-rank tests were performed;
4. p-values were converted into discriminability scores.

S7.3.1 Kaplan–Meier estimation

Kaplan–Meier survival estimation followed:

$$\hat{S}(t) = \prod_{t_i \le t} \left(1 - \frac{d_i}{n_i}\right)$$

where:

- $d_i$is event count,
- $n_i$is the number at risk.

The implementation used the lifelines survival-analysis library.

S7.3.2 Log-rank testing

Group survival differences were evaluated using the log-rank test.

The resulting p-value:

$$p_{\text{log-rank}}$$

was transformed into clinical discriminability:

$$D_{\text{clinical}} = -\log_{10}(p_{\text{log-rank}})$$

Larger D values correspond to stronger survival separation.

S7.4 Group-size and event-count constraints

To reduce unstable survival statistics, several quality-control thresholds were enforced.

Survival analysis was skipped if:

S7.4.1 Small patient groups

After median split:

$$n_{\text{group}} < 10$$

S7.4.2 Sparse event counts

Total event count:

$$E < 5$$

S7.4.3 Small cohort size

Total patient number:

$$N < 30$$

The formal implementation was:

These thresholds reduced unstable survival estimation arising from sparse clinical structure.

S7.5 KDE-based FWHM estimation

Observable width W was estimated using KDE-based FWHM analysis whenever possible.

The KDE workflow consisted of:

1. estimating pathway-expression density;
2. identifying maximum density;
3. computing the half-maximum threshold;
4. estimating the width of the density region above half maximum.

The formal implementation used:

gaussian_kde()

The resulting width:

$$W = x_{\text{right}} - x_{\text{left}}$$

represents the estimated observable cloud thickness.

S7.6 Fallback width estimation

If KDE estimation failed because of:

- sparse pathway size,
- numerical singularity,
- unstable density estimation,

an interquartile-range approximation was used.

The fallback approximation was:

$$W \approx 2.355 \times \frac{\mathrm{IQR}}{1.349}$$

The implementation was:

This fallback preserved numerical stability while maintaining approximate width scaling.

S7.7 Coherence-based pathway ranking

For K-scale analysis, pathway observables were ranked according to mean coherence:

$$C_i = \langle \frac{\mid \mu_i \mid}{W_i + \delta} \rangle$$

Pathways were sorted in descending order:

sort_values("coherence_mean", ascending=False)

The top K pathways were then selected for observation-scale analysis.

S7.8 λ-grid and K-grid configuration

The formal parameter grids used throughout D-PHY analyses were:

λ grid

$$\lambda \in \{0, 0.25, 0.50, 0.75, 1.0, 1.5, 2.0\}$$

defined as:

LAMBDA_GRID = [0, 0.25, 0.5, 0.75, 1.0, 1.5, 2.0]

K grid

$$K \in \{20, 50, 100, 200, 500\}$$

defined as:

K_VALUES = [20, 50, 100, 200, 500]

S7.9 Randomization reproducibility

Structured-versus-random analysis used repeated size-matched random sampling.

The formal random repetition number was:

$$N_{\text{random}} = 20$$

defined as:

N_RANDOM_PER_SET = 20

All random gene sets were processed using identical scoring and survival-analysis workflows.

S7.10 Cross-cancer independence

All analyses were performed independently for each cancer cohort.

Specifically:

- z-score normalization,
- pathway scoring,
- uncertainty estimation,
- λ optimization,
- K optimization,
- random-baseline generation

were recomputed separately within each cancer.

No pooled transcriptomic normalization across cancers was used.

This design preserved cancer-specific observable geometry.

S7.11 Visualization workflow

All figures were generated programmatically using scripted workflows.

The visualization framework included:

Small-multiple plots

Used for:

- D(K) curves,
- λ-dependent curves,
- cancer-wise comparisons.

Heatmaps

Used for:

- cross-cancer D(K) landscapes,
- K–λ interaction structure,
- pathway uncertainty organization.

Kaplan–Meier plots

Used for:

- representative top pathways,
- uncertainty-aware survival separation,
- structured-versus-random comparisons.

Scatter plots

Used for:

- K–λ coupling,
- coherence landscapes,
- pathway uncertainty distributions.

Structured-versus-random plots

Used for:

- empirical null comparisons,
- recurrence analysis,
- random-baseline visualization.

All plots were generated using scripted reproducible pipelines rather than manual figure editing.

S7.12 Output file organization

The pipeline generated multiple categories of outputs.

Summary tables

Pathway statistics

λ summaries

K summaries

Random-test summaries

Patient-level outputs

Patient scores

Survival groups

D_W-PHY values

Visualization outputs

PNG figures

PDF figures

KM plots

Heatmaps

Cross-cancer summaries

Optimal λ

Optimal K

Scale-regime classifications

Random recurrence tables

These outputs directly supported all main-text and supplementary figures.

S7.13 Reproducibility framework

The entire D-PHY workflow was designed as a reproducible protocol-based analysis system.

The reproducibility framework included:

1. fixed preprocessing procedures;
2. fixed pathway databases;
3. fixed λ grids;
4. fixed K grids;
5. fixed randomization structure;
6. standardized survival-analysis procedures;
7. scripted figure generation.

All major analyses were executed using reusable cohort-independent pipelines.

This protocol-level design was intended to reduce:

- manual intervention,
- hidden parameter tuning,
- cancer-specific workflow bias.

S7.14 Interpretation of statistical reproducibility within D-PHY

Within the D-PHY framework, reproducibility is interpreted not only as computational repeatability but also as stability of observable geometry under partial observability.

Accordingly:

Stable observables

are expected to exhibit:

- coherent uncertainty structure,
- reproducible discriminability,
- consistent scale behavior.

Unstable observables

are expected to exhibit:

- broad uncertainty clouds,
- inconsistent discriminability,
- unstable scale dependence.

Thus, statistical reproducibility and uncertainty geometry are treated as closely related properties within the D-PHY framework.

Supplementary Methods S8 | Figure generation, data integration workflow, and manuscript-specific output construction

This section describes the complete workflow used to generate manuscript figures, integrate cross-cancer outputs, construct summary visualization objects, and organize publication-ready outputs for the D-PHY analyses.

The procedures described here correspond to the formal figure-generation and result-integration pipelines used throughout AIDO-D-PHY I and D-PHY II.

S8.1 Conceptual rationale for figure construction

The D-PHY framework generates multiple layers of observable structure:

1. pathway-level uncertainty geometry;
2. patient-level uncertainty-aware discriminability;
3. cross-cancer uncertainty organization;
4. scale-dependent observation structure;
5. structured-versus-random contrasts.

Accordingly, figure generation was designed not merely for visualization, but for explicit representation of observable geometry under partial observability.

The visualization workflow therefore emphasized:

- cross-cancer comparability,
- uncertainty structure,
- observable landscapes,
- scale dependence,
- statistical reproducibility.

All figures were generated programmatically from exported summary tables to ensure reproducibility and avoid manual graphical modification.

S8.2 Output-directory organization

The formal workflow generated manuscript outputs into dedicated result directories.

The primary output structures included:

AIDO-D_RESULT_EXTRACTED_FOR_MANUSCRIPTS

AIDO-D_PHY_II_FIGURES

AIDO-D_RANDOM_TEST_RESULTS

The integrated summary tables were generated after all cancer-specific analyses completed.

The workflow therefore followed:

Cancer-level analysis → Summary-table export → Cross-cancer integration → Figure generation

S8.3 Exported summary-table structure

Each analysis stage generated structured CSV outputs.

The main exported categories included:

Pathway-level summaries

mu_mean_abs

W_mean

coherence_mean

mapped_gene_count

λ-scan summaries

Cancer

lambda

D_clinical

M_mean_abs_mu

U_mean_log_W

K-scale summaries

Cancer

K

lambda

D_clinical

Structured-versus-random summaries

Structured mean D

Structured max D

Random mean D

Random 95th percentile

Fraction above random baseline

High-D random recurrence summaries

Gene recurrence

Cross-cancer recurrence

High-D random overlap

These summary tables formed the direct input for all manuscript figures.

S8.4 Figure-generation architecture

The figure-generation framework was designed as a secondary scripted layer independent from the primary analysis pipeline.

Thus:

- analysis scripts generated summary CSV files,
- figure scripts loaded CSV summaries,
- visualization was reconstructed directly from exported data.

This separation improved:

- reproducibility,
- modularity,
- manuscript revision flexibility.

The figure-generation scripts therefore acted as reproducible rendering layers rather than primary analysis engines.

S8.5 Construction of uncertainty-landscape figures

Uncertainty-landscape figures visualized pathway observables in:

$$(\mu, W)$$

space.

Each point represented one pathway observable.

Axes typically represented:

- pathway activity magnitude,

- uncertainty width,
- coherence,
- discriminability.

The corresponding figure-generation workflow included:

1. loading pathway summary tables;
2. filtering invalid observations;
3. scaling axes consistently across cancers;
4. plotting pathway-level distributions.

These figures supported the interpretation of:

- state-cloud organization,
- uncertainty thickness,
- coherence landscapes.

S8.6 Construction of structured-versus-random figures

Structured-versus-random figures were generated from integrated random-test summaries.

The workflow consisted of:

1. loading structured summary statistics;
2. loading random-baseline statistics;
3. computing comparative metrics;
4. constructing grouped visualizations.

The principal visual outputs included:

- structured vs random bar plots,
- random-baseline distributions,
- recurrence histograms,
- cross-cancer comparison panels.

These figures visualize whether curated biological pathways exceed random transcriptomic organization.

S8.7 Construction of λ-scan figures

λ-scan figures were generated from patient-level uncertainty-aware analyses.

The workflow included:

1. loading λ-dependent discriminability summaries;
2. grouping by cancer cohort;
3. plotting:

$$D(\lambda)$$

curves.

For each cancer:

- λ values formed the x-axis,
- clinical discriminability formed the y-axis.

The maximal point:

$$\lambda^*$$

was highlighted using:

- vertical reference lines,
- point annotations,
- peak markers.

These visualizations supported uncertainty-sensitive discriminability interpretation.

S8.8 Construction of K-scale figures

K-scale figures visualized scale-dependent discriminability:

$$D(K)$$

and:

$$D(K, \lambda)$$

The workflow included:

1. loading K-scale summary tables;
2. grouping by cancer;
3. identifying maximal discriminability;
4. constructing:

- small-multiple plots,
- heatmaps,
- scale-regime summaries.

S8.8.1 Small-multiple D(K) plots

Each cancer cohort was visualized independently.

For each panel:

- x-axis:

$$K$$

- y-axis:

$$D(K)$$

The optimal observation scale:

$$K^*$$

was highlighted.

These figures visualize scale-dependent observational structure across cancers.

S8.8.2 Heatmap generation

Cross-cancer heatmaps were generated using matrices:

$$\mathbf{D} = [D(K, \lambda)]$$

Rows represented cancers.

Columns represented:

- K values,
- λ values,
- pathway databases.

Heatmap color represented discriminability magnitude.

These visualizations summarize global observable landscapes.

S8.9 Construction of K–λ interaction figures

K–λ interaction analysis examined the relationship between:

$$K^*$$

and:

$$\lambda^*$$

across cancers.

The workflow included:

1. identifying optimal K for each cancer;
2. identifying optimal λ for each cancer;
3. constructing:

- scatter plots,
- interaction heatmaps,
- correlation summaries.

The resulting figures visualize whether broader observational scales require stronger uncertainty regularization.

S8.10 Construction of scale-regime classification figures

Scale-regime figures summarized the morphology of:

$$D(K)$$

curves.

Cancers were classified into heuristic observational regimes:

- small-scale dominant,
- intermediate-optimal,
- large-scale dominant,
- flat/weak dependence.

The workflow included:

1. extracting optimal K;
2. comparing maximal D across scales;
3. assigning descriptive regime labels.

These classifications were exploratory and descriptive rather than formal statistical categories.

S8.11 Construction of Kaplan–Meier survival figures

Representative Kaplan–Meier plots were generated for:

- top discriminable pathways,

- uncertainty-aware patient scores,
- structured-versus-random comparisons.

The workflow included:

1. selecting representative observables;
2. performing median split;
3. estimating survival curves;
4. annotating:

- log-rank p-values,
- D values,
- pathway labels.

All Kaplan–Meier plots were generated directly from survival-analysis outputs.

S8.12 Figure export settings

All manuscript figures were exported programmatically.

Primary export formats included:

PNG

PDF

The workflow used consistent:

- DPI settings,
- axis formatting,
- label formatting,
- font scaling.

All figures were generated from script-based rendering pipelines rather than manual graphical editing.

S8.13 Cross-cancer integration workflow

Cross-cancer integration was performed after all cohort-specific analyses completed.

The workflow consisted of:

1. loading cohort-level outputs;
2. harmonizing variable names;
3. concatenating summary tables;

4. generating global comparison objects.

The integrated objects included:

$$\mathcal{D}_{\text{all cancers}}$$

for:

- D(K),
- D(\lambda),
- structured/random summaries,
- recurrence statistics.

This enabled direct cross-cancer visualization and regime comparison.

S8.14 Reproducibility safeguards for figure generation

Several safeguards were implemented during figure generation.

Script-based rendering

All figures generated from exported summary tables.

Fixed parameter grids

The same:

- λ values,
- K values,
- pathway databases

were used across all figures.

Independent cancer processing

Figures were generated from independently processed cohorts.

Automated peak identification

Optimal:

$$K^*$$

and:

$$\lambda^*$$

were identified programmatically.

Consistent axis scaling

Cross-cancer comparisons used harmonized visualization scales where appropriate.

S8.15 Biological interpretation of figure-generation architecture

Within the D-PHY framework, visualization is treated as part of observable-geometry reconstruction rather than merely graphical presentation.

Thus:

Heatmaps

represent:

- cross-cancer observable landscapes.

D(K) curves

represent:

- observation-scale response functions.

K–λ interaction figures

represent:

- coupling between observational coverage and uncertainty regulation.

Structured-versus-random figures

represent:

- comparison between biological organization and baseline statistical geometry.

Accordingly, the figure architecture itself reflects the conceptual structure of uncertainty-aware observable analysis under partial observability.

S8.16 Output objects generated from figure-generation workflows

The final figure-generation pipeline produced:

Publication figures

Main figures

Supplementary figures

Cross-cancer summary panels

Integrated summary tables

K-scale summary tables

λ summary tables

Random-test summary tables

Reproducibility outputs

Figure-generation scripts

Processed summary CSV files

Visualization metadata

These outputs collectively formed the publication-ready representation of the D-PHY framework.

Supplementary Methods S9 | Biological interpretation framework, partial observability formulation, and uncertainty-geometry representation

This section describes the conceptual interpretation framework underlying the D-PHY analyses. Unlike conventional pathway-scoring pipelines that treat transcriptomic observables primarily as descriptive molecular features, the D-PHY framework interprets transcriptomic pathway observables as uncertainty-bearing projections generated under partial observability.

The objective of this section is to formally define the interpretation layer connecting:

- transcriptomic observations,
- pathway-level uncertainty structure,
- discriminability,
- observable geometry,
- partial observability.

This interpretation framework provides the conceptual basis for the μ–W representation, uncertainty-aware discriminability analysis, and scale-dependent observation analysis developed in the main manuscript.

S9.1 Partial observability in transcriptomic cancer systems

Bulk transcriptomic measurements do not provide direct access to complete latent biological organization.

Observed gene-expression values arise from multiple unresolved contributors, including:

- tumor-cell populations,
- stromal mixtures,
- immune infiltration,
- clonal heterogeneity,
- signaling interactions,
- epigenetic regulation,
- microenvironmental coupling.

Consequently, transcriptomic observations are interpreted as effective projected measurements rather than direct state readouts.

Within the D-PHY framework, transcriptomic observation is represented as:

$$O = h(S) + \epsilon$$

where:

- S denotes latent biological organization,
- $h(\cdot)$ denotes the effective observation mapping,
- O denotes transcriptomic observables,
- ϵ represents unresolved variability and measurement uncertainty.

Under this formulation:

- multiple latent configurations may produce similar transcriptomic observations,
- observable similarity does not necessarily imply latent-state equivalence.

This forms the basis of partial observability within the D-PHY framework.

S9.2 Biological-process observables as projected representations

Pathway-level transcriptomic observables are treated as structured projections of the latent system.

For biological process B_i:

$$O_{B_i} = h_{B_i}(S) + \epsilon_{B_i}$$

where:

- h_{B_i} denotes the pathway-specific projection mapping,
- O_{B_i} denotes the pathway-level observable,

- ϵ_{B_i} denotes pathway-specific uncertainty.

Thus, pathway observables are interpreted as:

- compressed representations,
- uncertainty-bearing projections,
- partial summaries of latent organization.

This differs from interpretations that treat pathway scores as direct mechanistic readouts.

S9.3 State-cloud interpretation of pathway observables

Within the D-PHY framework, pathway observables are represented geometrically as uncertainty clouds.

For pathway B_i, the observable cloud is represented by:

$$(\mu_i, W_i)$$

where:

- μ_i denotes cloud center,
- W_i denotes cloud thickness.

S9.3.1 Observable center

$$\mu_i$$

represents the effective activity location of the pathway observable.

Operationally, this corresponds to average pathway activation measured from transcriptomic projection.

S9.3.2 Observable width

$$W_i$$

represents pathway uncertainty thickness estimated from transcriptomic dispersion.

Broad width indicates:

- heterogeneous observable organization,
- projection overlap,
- unresolved latent variability.

Narrow width indicates:

- concentrated observable organization,
- reduced projection ambiguity.

This interpretation forms the core μ–W state-cloud representation.

S9.4 Interpretation of FWHM as uncertainty thickness

The full width at half maximum:

$$W = \mathrm{FWHM}$$

was used as an operational uncertainty-width estimate.

Within the D-PHY framework:

$$W$$

does not represent measurement noise alone.

Instead, it represents effective uncertainty arising from:

- latent-state multiplicity,
- projection compression,
- pathway overlap,
- unresolved biological heterogeneity,
- transcriptomic mixing effects.

Thus, FWHM is interpreted as:

operational state-cloud thickness.

This interpretation connects pathway-level transcriptomic structure to uncertainty geometry under partial observability.

S9.5 Observable coherence as uncertainty compression

Observable coherence was defined as:

$$C_i = \frac{|\mu_i|}{W_i + \delta}$$

Within the D-PHY framework:

- coherence measures effective uncertainty compression,
- high coherence indicates concentrated observable organization,

- low coherence indicates diffuse observable geometry.

Importantly, coherence is not interpreted simply as signal-to-noise ratio.

Instead, coherence represents the balance between:

1. observable activity magnitude,
2. uncertainty expansion under projection.

Thus:

High coherence

suggests that pathway organization remains relatively preserved under transcriptomic projection.

Low coherence

suggests substantial uncertainty expansion and projection overlap.

S9.6 Interpretation of uncertainty-aware discriminability

Clinical discriminability was defined operationally as:

$$D_{\text{clinical}} = -\log_{10}(p_{\text{log-rank}})$$

Within the D-PHY framework, discriminability is interpreted as a measure of how effectively observable structure preserves clinically relevant separation under partial observability.

Thus:

High discriminability

emerges when:

- observable organization is coherent,
- uncertainty overlap is limited,
- clinically relevant latent variation remains preserved.

Low discriminability

emerges when:

- observable clouds strongly overlap,
- projection ambiguity dominates,
- latent-state information becomes compressed or diluted.

Consequently, discriminability is interpreted as an emergent property of observable geometry rather than merely a statistical endpoint.

S9.7 Interpretation of uncertainty-aware scoring

The patient-level uncertainty-aware score:

$$D_{W\text{-}PHY,p} = M_p - \lambda U_p$$

combines:

- observable magnitude,
- uncertainty burden.

Within the D-PHY framework:

$$\boldsymbol{M_p}$$

represents average pathway activation magnitude.

$$\boldsymbol{U_p}$$

represents uncertainty accumulation across pathway observables.

λ

controls uncertainty penalization strength.

Thus, uncertainty-aware scoring evaluates whether clinically useful transcriptomic organization emerges preferentially from:

- concentrated observable structure,
- reduced uncertainty overlap.

S9.8 Interpretation of λ

The λ parameter does not represent a physical constant.

Instead, λ acts as an operational uncertainty-weighting parameter controlling how strongly pathway uncertainty contributes to patient-level scoring.

Small λ

emphasizes pathway activity magnitude.

Large λ

strongly penalizes broad uncertainty structure.

Optimal λ

represents the balance between:

- preserving biological activity,
- suppressing uncertainty accumulation.

Distinct cancers exhibited distinct:

$$\lambda^*$$

values, suggesting heterogeneous uncertainty organization across cancer systems.

S9.9 Scale-dependent observation interpretation

Within the D-PHY framework, observation scale:

$$K$$

represents observational coverage.

Small K corresponds to highly selective observations.

Large K corresponds to broader observational integration.

Importantly:

$$D(K)$$

was not assumed to increase monotonically.

Increasing observation scale may simultaneously:

- improve latent-state coverage,
- increase projection overlap,
- increase uncertainty accumulation,
- dilute coherent observable organization.

Thus:

Optimal K

emerges from the balance between:

1. informational coverage,
2. uncertainty compression.

S9.10 Interpretation of structured-versus-random observables

The structured-versus-random framework evaluates whether biologically curated pathways preserve additional organization beyond generic high-dimensional transcriptomic geometry.

Within the D-PHY framework:

Random gene sets

represent baseline statistical geometry of transcriptomic space.

Structured pathways

represent biologically constrained reshaping of this geometry.

Thus, high discriminability in structured pathways reflects:

- uncertainty compression,
- biologically organized observable structure,
- preservation of clinically relevant latent variation.

S9.11 Biological interpretation of high-D random observables

The presence of high-discriminability random gene sets does not imply absence of biological structure.

Instead, high-D random observables suggest that:

- transcriptomic spaces contain intrinsic latent organization,
- biologically meaningful axes may partially emerge even under random sampling.

This observation motivated the high-D random dissection analysis and cross-cancer recurrence framework.

However, these analyses were treated as exploratory rather than definitive mechanistic evidence.

S9.12 Cross-cancer uncertainty landscapes

Different cancers exhibited distinct:

- coherence distributions,
- λ responses,
- K-dependent discriminability curves,
- uncertainty geometries.

Within the D-PHY framework, these differences are interpreted as cancer-specific observable landscapes arising from distinct latent organizational structures.

Thus, transcriptomic observability itself is treated as cancer dependent.

S9.13 Relationship between observable geometry and clinical separability

The D-PHY framework proposes that clinically meaningful transcriptomic observables emerge when:

1. pathway activity remains sufficiently organized,
2. uncertainty expansion remains sufficiently constrained,
3. observable overlap remains limited.

Accordingly, clinical separability is interpreted as a manifestation of observable geometry under projection constraints.

This differs from conventional interpretations that treat survival separation purely as a statistical association problem.

S9.14 Interpretation of D-PHY as an uncertainty-geometry framework

The D-PHY framework does not attempt to reconstruct unique latent biological states directly from transcriptomic observations.

Instead, it evaluates:

- observable uncertainty structure,
- uncertainty compression,
- clinically relevant observable organization.

Thus, D-PHY is fundamentally an operational observable-geometry framework under partial observability.

The framework therefore focuses on:

- pathway-level uncertainty geometry,
- discriminability landscapes,
- observational scale structure,
- uncertainty-aware projection organization.

rather than deterministic latent-state reconstruction.

S9.15 Summary interpretation framework

The complete D-PHY interpretation layer can be summarized as:

$$\text{Latent organization} \rightarrow \text{Transcriptomic projection} \rightarrow (\mu, W)$$
$$\rightarrow \text{Observable geometry} \rightarrow D_{\text{clinical}}$$

Within this framework:

- μrepresents observable center,
- Wrepresents uncertainty thickness,
- coherence represents uncertainty compression,
- discriminability represents clinically relevant observable separation.

Thus, transcriptomic pathway observables are interpreted as structured uncertainty-bearing projections generated under partial observability.

Supplementary Methods S10 | Result interpretation framework, ranking strategy, evidence hierarchy, and methodological limitations

This section describes the interpretation framework used throughout the D-PHY analyses. The purpose of this section is to clarify how discriminability values, uncertainty structures, pathway rankings, and cross-cancer comparisons were interpreted, as well as to distinguish between primary evidence and exploratory analyses.

Because the D-PHY framework combines:

- pathway-level transcriptomic observables,
- uncertainty-width estimation,
- survival-based discriminability,
- scale-dependent analysis,
- structured-versus-random comparisons,

explicit interpretation constraints are necessary to avoid overinterpretation of operational observables as direct biological-state measurements.

The procedures described here were applied consistently throughout the analysis and manuscript interpretation process.

S10.1 Interpretation of clinical discriminability

Clinical discriminability was defined as:

$$D_{\text{clinical}} = -\log_{10}(p_{\text{log-rank}})$$

where:

- $p_{\text{log-rank}}$ denotes the log-rank test p-value obtained from Kaplan–Meier survival analysis.

Within this formulation:

Larger D values

indicate:

- stronger survival separation,
- higher observable-associated outcome divergence,
- stronger operational clinical discriminability.

S10.1.1 Operational interpretation ranges

For practical interpretation, D values were interpreted approximately as:

D value	Approximate interpretation
$D < 1$	weak discriminability
$1 \leq D < 2$	moderate discriminability
$2 \leq D < 3$	strong discriminability
$D \geq 3$	highly discriminable observable

These ranges were used descriptively rather than as rigid statistical thresholds. Importantly, D values were interpreted comparatively within the same analysis framework rather than as absolute biological constants.

S10.2 Interpretation of uncertainty width W

Within the D-PHY framework:

$$W = \mathrm{FWHM}$$

was interpreted as an operational uncertainty-width estimate.

Importantly:

$$W$$

does not represent:

- thermodynamic entropy,
- direct physical disorder,
- literal latent-state multiplicity.

Instead, W represents an operational observable-width proxy reflecting:

- projection dispersion,
- pathway overlap,
- transcriptomic heterogeneity,
- unresolved latent variability,
- observable compression effects.

Thus:

Narrow W

suggests concentrated observable organization.

Broad W

suggests diffuse observable geometry and increased projection ambiguity.

S10.3 Interpretation of coherence

Observable coherence was defined as:

$$C = \frac{|\mu|}{W + \delta}$$

Within the D-PHY framework:

- coherence measures uncertainty compression,
- not signal-to-noise ratio alone.

High coherence was interpreted as:

- concentrated observable geometry,
- reduced uncertainty spread,
- preserved transcriptomic organization.

Low coherence was interpreted as:

- broad observable overlap,
- uncertainty expansion,
- diffuse transcriptomic organization.

S10.4 Interpretation of uncertainty-aware scoring

The uncertainty-aware patient score:

$$D_{W\text{-}PHY,p} = M_p - \lambda U_p$$

was interpreted as an operational observable score rather than a literal patient "disease state."

Importantly:

$$D_{W\text{-}PHY,p}$$

was used only for:

- patient stratification,
- survival separation,
- discriminability analysis.

It was not interpreted as:

- causal disease severity,
- latent-state reconstruction,
- direct biological progression measurement.

S10.5 Interpretation of λ

The parameter:

$$\lambda$$

was treated as an operational uncertainty-weighting parameter.

It does not represent:

- a universal physical constant,
- a biological kinetic parameter,
- a mechanistic tumor property.

Instead, λ controls how strongly uncertainty width contributes to pathway scoring.

Small λ

emphasizes pathway activity magnitude.

Large λ

strongly penalizes broad uncertainty structure.

Optimal λ

represents the uncertainty weighting producing maximal operational discriminability within the current analysis framework.

Distinct cancers may therefore exhibit distinct:

$$\lambda^*$$

values because of differences in observable organization and transcriptomic geometry.

S10.6 Interpretation of observation scale K

Observation scale:

$$K$$

represents the number of pathway observables included in uncertainty-aware scoring.

Importantly:

$$K$$

does not represent:

- biological system size,
- pathway complexity,
- latent-state dimensionality directly.

Instead, K operationally represents observational coverage.

Small K

corresponds to highly selective observation.

Large K

corresponds to broader transcriptomic integration.

Optimal K

represents the observational scale producing maximal discriminability under the current framework.

The resulting:

$$K^*$$

therefore reflects effective observational organization rather than direct biological dimensionality.

S10.7 Primary evidence versus exploratory evidence

Because the D-PHY framework contains multiple analysis layers, analyses were classified into:

1. primary evidence;
2. exploratory evidence.

This distinction was used throughout manuscript interpretation.

S10.7.1 Primary evidence

The following analyses were treated as primary evidence:

Structured-versus-random comparison

Used to evaluate whether curated pathways exceed random transcriptomic organization.

λ-scan analysis

Used to evaluate uncertainty-sensitive discriminability.

D(K) analysis

Used to evaluate scale-dependent observable structure.

Representative survival separation

Used to demonstrate operational clinical discriminability.

Cross-cancer uncertainty landscapes

Used to demonstrate heterogeneous observable organization across cancers.

D_μ

These analyses directly support the primary claims of the manuscript.

S10.7.2 Exploratory evidence

The following analyses were treated as exploratory:

High-D random recurrence

Cross-cancer recurrent random genes

K-regime classification

K–λ coupling interpretation

Latent geometry interpretation

These analyses were interpreted cautiously and were not treated as definitive mechanistic evidence.

S10.8 Representative pathway selection strategy

Representative pathways used in Kaplan–Meier plots and manuscript figures were selected according to:

1. high clinical discriminability;
2. reproducibility across analyses;
3. biologically interpretable pathway identity;
4. stable survival separation.

Representative pathways were not selected manually based solely on visual appearance.

Pathway selection was performed programmatically using ranking tables generated from:

$$D_\mu$$

$$D_W$$

or:

$$D_{W\text{-}PHY}$$

summaries.

S10.9 Cross-cancer interpretation constraints

Direct comparison of absolute D values across cancers was interpreted cautiously.

Differences between cancers may arise from:

- cohort size,
- event rate,
- survival structure,
- transcriptomic heterogeneity,
- observable geometry.

Therefore:

Within-cancer comparisons

were considered more reliable than direct absolute comparisons across cancers.

Cross-cancer analyses were primarily interpreted qualitatively in terms of:

- observable organization,
- scale dependence,
- uncertainty structure.

S10.10 Interpretation of random observables

The presence of high-discriminability random gene sets was interpreted cautiously.

High-D random observables do not imply absence of biological structure.

Instead, they suggest that:

- high-dimensional transcriptomic spaces possess intrinsic statistical geometry,
- biologically relevant latent axes may partially emerge under random sampling.

Therefore:

Structured pathways

were interpreted as biologically constrained reshaping of baseline transcriptomic geometry.

S10.11 Why operational uncertainty proxies were used

The D-PHY framework uses operational uncertainty proxies rather than direct latent-state entropy measurements.

This design was necessary because:

- latent biological states are inaccessible,
- true state distributions are unknown,
- transcriptomic observations are partial projections.

Consequently:

$$W$$

$$C$$

$$D_{W\text{-}PHY}$$

were interpreted as operational observability proxies rather than direct physical-state measurements.

This distinction is essential for avoiding overinterpretation of transcriptomic observables.

S10.12 Interpretation limitations

Several important limitations apply to the D-PHY framework.

Survival is not direct latent-state evolution

Clinical survival endpoints do not directly measure:

$$S(t) \to S(t+1)$$

latent-state trajectories.

Instead, survival acts as an operational downstream outcome proxy.

Pathway activity is not direct mechanism

Pathway scores represent transcriptomic projections rather than mechanistic pathway activation measurements.

W is not literal entropy

FWHM-based width is an operational uncertainty-width estimate rather than thermodynamic entropy.

D is not causality

High discriminability indicates outcome-associated separation but does not imply causal biological mechanisms.

K and λ are operational parameters

Optimal:

$$K^*$$

and:

$$\lambda^*$$

depend on the present analysis framework and should not be interpreted as universal biological constants.

S10.13 Interpretation of observable geometry under partial observability

Within the D-PHY framework, transcriptomic pathway observables are interpreted geometrically.

Observable center

$$\mu$$

represents pathway activity location.

Observable width

$$W$$

represents uncertainty thickness.

Coherence

$$C$$

represents uncertainty compression.

Discriminability

$$D$$

represents clinically relevant observable separation.

Thus, observable quality emerges from the geometry of uncertainty-bearing transcriptomic projections rather than from pathway activity magnitude alone.

S10.14 Evidence architecture of manuscript figures

The manuscript figures were organized according to evidence hierarchy.

Figure 1

Conceptual framework and uncertainty-geometry representation.

Figure 2

Cross-cancer uncertainty landscapes and pathway geometry.

Figure 3

Structured-versus-random validation.

Figure 4

λ-dependent uncertainty-aware discriminability.

Figure 5

Scale-dependent discriminability and K-scale organization.

Figure 6

Representative pathway survival separation.

This architecture separates:

- conceptual framework,
- primary evidence,

- exploratory interpretation,
- representative biological examples.

S10.15 Summary interpretation framework

The D-PHY interpretation framework can be summarized as:

Transcriptomic projection → (μ, W) → Observable geometry
→ Uncertainty-aware discriminability → Clinical separability

Within this framework:

- pathway observables are uncertainty-bearing projections,
- discriminability is an operational observable-separation metric,
- uncertainty geometry shapes observable usefulness under partial observability.

This interpretation framework underlies all analyses reported in the present study.

Supplementary Methods S11 | Computational reproducibility framework, execution protocol, code organization, and data availability

This section describes the complete computational reproducibility framework used throughout the D-PHY analyses, including execution order, script organization, directory structure, exported intermediate objects, figure-regeneration procedures, and reproducibility safeguards.

The objective of this section is to provide a protocol-level description sufficient for independent reconstruction of the analytical workflow and manuscript figures.

The D-PHY framework was implemented as a reusable pipeline architecture rather than a manually tuned cancer-specific workflow.

S11.1 Overall execution architecture

The complete D-PHY workflow consisted of four major computational layers:

Layer 1 | Data preprocessing

This layer performs:

- transcriptomic loading,
- survival-table parsing,
- TCGA barcode harmonization,
- z-score normalization,
- sample intersection.

Output:

$$Z$$

normalized transcriptomic matrices.

Layer 2 | Pathway observable construction

This layer performs:

- GMT parsing,
- pathway mapping,
- pathway activity computation,
- uncertainty-width estimation,
- coherence calculation.

Outputs:

$$\mu$$

and:

$$W$$

matrices.

Layer 3 | Discriminability analysis

This layer performs:

- λ-scan analysis,
- K-scale analysis,
- Kaplan–Meier survival analysis,
- random-baseline generation,
- structured-versus-random comparison.

Outputs:

- D values,
- optimal λ,

- optimal K,
- random statistics.

Layer 4 | Figure-generation and integration

This layer performs:

- summary-table integration,
- cross-cancer aggregation,
- heatmap generation,
- D(K) curve construction,
- manuscript figure rendering.

Outputs:

- PNG figures,
- PDF figures,
- publication-ready summary tables.

S11.2 Formal execution order

The formal execution order of the D-PHY workflow was:

Step 1 | TCGA preprocessing

Input:

Expression matrix

Survival table

Output:

Normalized expression matrix

Matched patient list

Step 2 | Pathway database loading

Input:

GMT files

Output:

Mapped pathway sets

Step 3 | Pathway observable generation

Output:

μ matrix

W matrix

Coherence table

Step 4 | λ-scan analysis

Output:

D(λ)

Optimal λ*

Step 5 | K-scale analysis

Output:

D(K)

Optimal K*

K–λ summary

Step 6 | Structured-versus-random analysis

Output:

Structured/random summaries

Random baselines

High-D random tables

Step 7 | Cross-cancer integration

Output:

Integrated summary CSV files

Step 8 | Figure generation

Output:

Publication figures

Supplementary figures

S11.3 Input-file structure

The pipeline requires three primary categories of input files.

S11.3.1 Transcriptomic expression files

Supported expression filenames included:

GE.tsv

HiSeqV2.tsv

expression.tsv

Rows represent genes.

Columns represent patient samples.

S11.3.2 Clinical survival files

Supported survival filenames included:

survival.tsv

Phenotype.tsv

clinical.tsv

The pipeline automatically detects:

- survival-time columns,
- event-status columns.

S11.3.3 Pathway GMT files

Supported GMT pathway collections included:

Hallmark

GO_BP

Reactome

KEGG_Medicus

WikiPathways

S11.4 Directory organization

The workflow used standardized directory organization.

Raw-data directories

TCGA/

GMT/

Intermediate outputs

SUMMARY/

MATRICES/

TEMP/

Final outputs

FIGURES/

MANUSCRIPT_EXPORT/

Integrated manuscript summaries

AIDO-D_RESULT_EXTRACTED_FOR_MANUSCRIPTS

This structure separates:

- raw data,
- intermediate objects,
- publication outputs.

S11.5 Script categories

The workflow was organized into modular script categories.

Preprocessing scripts

Responsibilities:

- TCGA loading,
- barcode matching,
- normalization.

Outputs:

Normalized matrices

QC summaries

Pathway-construction scripts

Responsibilities:

- GMT parsing,
- pathway mapping,
- μ/W computation,
- coherence calculation.

Outputs:

μ matrix

W matrix

Pathway summaries

λ-analysis scripts

Responsibilities:

- λ-grid scanning,
- patient scoring,
- survival analysis.

Outputs:

D(λ)

Optimal λ

K-scale scripts

Responsibilities:

- pathway ranking,
- top-K selection,
- D(K) analysis.

Outputs:

K-scale summaries

K* tables

Random-test scripts

Responsibilities:

- random gene-set generation,
- structured/random comparison,

- recurrence analysis.

Outputs:

Random baseline tables

High-D recurrence summaries

Figure-generation scripts

Responsibilities:

- CSV loading,
- cross-cancer integration,
- figure rendering.

Outputs:

PNG figures

PDF figures

S11.6 Script-to-output mapping

The major figure-generation workflow followed:

Analysis layer	Primary output
Pathway construction	μ/W summary tables
λ scan	D(λ) tables
K-scale analysis	D(K) tables
Random analysis	structured/random summaries
Cross-cancer integration	manuscript CSV summaries
Figure rendering	publication figures

Thus, figures were reconstructed from exported summaries rather than directly from raw expression matrices.

S11.7 Reproducibility safeguards

Several reproducibility safeguards were implemented.

Standardized preprocessing

All cancers used:

- identical normalization,
- identical pathway-processing logic,
- identical survival-analysis procedures.

Fixed parameter grids

The same:

$$\lambda$$

and:

$$K$$

grids were used across all cancers.

Independent cohort processing

Each cancer was analyzed independently.

No pooled cross-cancer normalization was performed.

Shared survival procedures

All survival analyses used:

- median split,
- Kaplan–Meier estimation,
- log-rank testing.

Shared randomization framework

All structured-versus-random analyses used:

- identical pathway-size matching,
- identical random repetition counts.

S11.8 Figure-regeneration protocol

All manuscript figures can be regenerated directly from exported summary CSV files.

The figure-generation workflow does not require rerunning:

- transcriptomic preprocessing,
- pathway scoring,

- survival analysis.

Instead:

Summary CSV

→

Figure-generation script

→

Publication figure

This separation improves:

- reproducibility,
- manuscript revision efficiency,
- visualization consistency.

S11.9 Cross-cancer integration workflow

After all cancer-specific analyses completed:

1. summary CSV files were collected;
2. column structures were harmonized;
3. integrated tables were generated.

The integrated summary tables included:

D(λ)

D(K)

K*

λ*

Structured/random statistics

These integrated tables formed the direct input for:

- heatmaps,
- D(K) small multiples,
- K–λ interaction plots,
- scale-regime summaries.

S11.10 Figure export and rendering settings

All figures were generated programmatically using:

- matplotlib,

- pandas-based plotting workflows.

Primary export formats included:

PNG

PDF

Figures used consistent:

- DPI,
- font scaling,
- axis formatting,
- annotation styles.

No manual graphical editing was required to regenerate the figures.

S11.11 Runtime behavior and fallback procedures

Several fallback procedures were implemented to maintain numerical stability.

KDE failure fallback

If KDE-based FWHM estimation failed:

- IQR-based width approximation was used.

Missing survival handling

Samples lacking:

- survival time,
- event status

were excluded automatically.

Sparse-group handling

Survival analysis skipped if:

$$n_{\text{group}} < 10$$

or:

$$E < 5$$

Duplicate handling

Duplicated genes and duplicated patient identifiers were collapsed automatically.

These safeguards reduced instability during large-scale cross-cancer processing.

S11.12 Data availability

The D-PHY framework used publicly available TCGA transcriptomic and clinical datasets obtained from UCSC Xena.

The pathway databases were obtained from publicly available GMT pathway repositories compatible with MSigDB/GSEA workflows.

Processed summary tables generated during analysis included:

μ summaries

W summaries

D(λ)

D(K)

Structured/random summaries

These processed outputs are sufficient for figure regeneration and downstream reproducibility analysis.

S11.13 Code availability

All major analyses were implemented using scripted Python workflows.

The reproducibility framework included:

- preprocessing scripts,
- pathway-construction scripts,
- λ-scan scripts,
- K-scale scripts,
- random-test scripts,
- figure-generation scripts.

The workflow was designed as a reusable protocol-level framework rather than a manually tuned cohort-specific analysis.

Code availability and repository release follow journal and repository policies.

S11.14 Computational limitations

Several computational limitations apply to the current workflow.

KDE sensitivity

KDE-based FWHM estimation may become unstable for:

- extremely sparse pathways,
- narrow distributions.

Survival sparsity

Certain cancers exhibit:

- low event counts,
- limited survival separation.

Pathway-size effects

Extremely small or large pathways may produce unstable uncertainty estimates.

Cohort imbalance

Different cancers exhibit different:

- sample sizes,
- survival structures,
- transcriptomic heterogeneity.

Thus, cross-cancer interpretation should be performed cautiously.

S11.15 Reproducibility philosophy of the D-PHY framework

The D-PHY framework was designed as:

a protocol-level reproducible uncertainty-aware transcriptomic analysis system.

The objective was to minimize:

- manual parameter tuning,
- cancer-specific workflow modifications,
- ad hoc figure generation.

Instead, the framework emphasizes:

- reusable execution pipelines,
- modular analysis layers,
- summary-table-based integration,
- programmatic figure regeneration.

This reproducibility architecture forms a central component of the D-PHY framework.

Supplementary Information Summary

Together, SM1–SM11 form a reproducible uncertainty-aware transcriptomic analysis framework designed to evaluate biological-process observables under partial observability. The supplementary materials provide both the computational implementation details and the conceptual interpretation layer required to reconstruct the D-PHY analytical workflow and reproduce the manuscript results.

The complete workflow can be summarized as:

Transcriptomic preprocessing

→ Pathway-level observable construction

→ Uncertainty-width estimation

→ Uncertainty-aware discriminability analysis

→ Scale-dependent observation analysis

→ Structured-versus-random validation

→ Cross-cancer integration

→ Reproducible figure generation

Within this framework, transcriptomic pathway observables are interpreted not merely as descriptive molecular features, but as uncertainty-bearing projections generated under partial observability. The supplementary materials therefore serve both as a protocol-level reproducibility document and as an extended methodological foundation for the D-PHY framework.